SIMULTANEOUS VISIBILITY IN CUT-AND-PROJECT SETS

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Abstract. Visibility from the origin in integer lattices is a well-studied problem. More recently, the corresponding question for cut-and-project sets has attracted attention through the work of Marklof and Strömbergsson, Hammarhjelm, and Gringlaz-Kumar-Weiss. Simultaneous visibility asks which points in a set are visible from every point of another set. This question was studied for the integer lattice by Rearick and Rumsey, among others.

In this work, we investigate simultaneous visibility in cut-and-project sets associated with real quadratic fields satisfying the Hammarhjelm condition. Under suitable geometric conditions on the finite set, we prove that the density of the simultaneously visible points exists and derive an explicit formula for it. We also examine simultaneous visibility in more general cut-and-project sets under a condition on the associated lattice.

1. Introduction

Let $\Lambda \subset \mathbb{R}^d$ be a locally finite point set, and let $S \subset \mathbb{R}^d$ be a set. We denote by $\Lambda_{\mathrm{vis},S}$ the set of points in $\Lambda \backslash S$ that are visible from every point $\boldsymbol{s} \in S$. Geometrically, these are the points that can be reached simultaneously from all points of S by straight line segments whose interiors do not pass through any other point of Λ:

$$\Lambda_{\mathrm{vis},S} = \{\boldsymbol{y} \in \Lambda \backslash S : \forall \boldsymbol{s} \in S,\ \forall t \in (0,1),\ t\boldsymbol{s} + (1-t)\boldsymbol{y} \notin \Lambda\}\,.$$

Let $\mathcal{A} = \{A_T\}_{T \in \mathcal{I}}$ be an increasing family of sets whose union is $\mathbb{R}^d$, where $\mathcal{I} \subset (0,\infty)$ is unbounded. The asymptotic density of Λ along $\mathcal{A}$ is defined by

$$\theta(\Lambda) \overset{\text{def}}{=} \lim_{T \to \infty} \frac{\#(\Lambda \cap A_T)}{\mathrm{vol}(A_T)},$$

provided the limit exists.

The integer lattice $\Lambda = \mathbb{Z}^k$, with $k \geqslant 2$, is the classical example. The set $\mathbb{Z}^k_{\mathrm{vis},\{\mathbf{0}\}}$ of lattice points visible from the origin is usually called the set of primitive points. It is well known that its density is

$$\theta\left(\mathbb{Z}^k_{\mathrm{vis},\{\mathbf{0}\}}\right) = \frac{1}{\zeta(k)}$$

for standard families of sets $\mathcal{A}$ such as cubes and balls. See for example [BMP00] and its appendix for a discussion of more general families. It was shown in [Erd58] that the set $\mathbb{Z}^k$ contains arbitrarily large cubes containing no visible points from the origin (see also [BMP00, Proposition 4]).

The set $\mathbb{Z}^k_{\mathrm{vis},\{\mathbf{0}\}}$ is uniformly discrete but it is not relatively dense and hence is not a Delone set (see Section 2.2 for the definition). In fact, since these holes repeat periodically throughout the lattice, adjoining any set of zero density to $\mathbb{Z}^k_{\mathrm{vis},\{\mathbf{0}\}}$ still does not produce a Delone set.

1.1. Simultaneous visibility in the integer lattice.

Denote by $\mathcal{P}$ the set of prime numbers and let $S \subset \mathbb{Z}^k$. For each $p \in \mathcal{P}$, let

$$\pi_p : \mathbb{Z}^k \longrightarrow \left(\mathbb{Z}/p\mathbb{Z}\right)^k, \qquad (x_1, \ldots, x_k) \longmapsto (x_1 \bmod p, \ldots, x_k \bmod p)$$

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be the coordinatewise projection. We write

$$s(p) \stackrel{\text{def}}{=} \#(\pi_p(S))$$

for the number of distinct residue classes modulo p represented by the elements of S.

Rumsey [Rum66, Theorem 7] proved that, with respect to the family of cubes $[-n,n]^k$, the density of $\mathbb{Z}^k_{\text{vis},S}$ exists and is given by

$$\theta\left(\mathbb{Z}^k_{\text{vis},S}\right) = \prod_{p\in\mathcal{P}} \left(1 - \frac{s(p)}{p^k}\right).$$

The special case in which S is finite and its points are pairwise visible had been studied earlier by Rearick [Rea66], who established the same formula with an effective error term. Rumsey observed that the same density formula holds for a family of boxes

$$\{[-n_1,n_1]\times\cdots\times[-n_k,n_k] : \min\{n_1,\ldots,n_k\}\to\infty\},$$

and extended the formula for certain infinite subsets S satisfying appropriate hypotheses. For an arbitrary finite set S, density along a family of Euclidean balls follows from [KT26, Theorem 1.3]. We refer the reader to [BKP23; ACZ23; PH13; BMP00; Ara26] and the references therein for further details on simultaneous visibility and related results.

1.2. **Simultaneous visibility in cut-and-project sets.** Let $n = d+m$, where $d, m \in \{1,2,\ldots\}$, and consider the decomposition

$$\mathbb{R}^n = \mathbb{R}^d \oplus \mathbb{R}^m. \tag{1}$$

Let $\mathcal{L}$ be a lattice in $\mathbb{R}^n$, and denote by π_{phys} and π_{int} the projections onto the first d coordinates and the last m coordinates, respectively. Let $\mathcal{W}\subset\mathbb{R}^m$ be a bounded measurable subset, called the *window*. We assume that $\pi_{\text{int}}(\mathcal{L})$ is dense in $\mathbb{R}^m$, $\pi_{\text{phys}}|_{\mathcal{L}}$ is injective. The associated cut-and-project set is defined by

$$\Lambda(\mathcal{W},\mathcal{L}) \stackrel{\text{def}}{=} \{\pi_{\text{phys}}(\boldsymbol{x}) : \boldsymbol{x}\in\mathcal{L},\ \pi_{\text{int}}(\boldsymbol{x})\in\mathcal{W}\}. \tag{2}$$

We say that $\Lambda = \Lambda(\mathcal{W},\mathcal{L})$ is *irreducible* if the window $\mathcal{W}$ is bounded, has nonempty interior, and has boundary of Lebesgue measure zero. Let $D\subset\mathbb{R}^d$ be a Jordan measurable set with $\text{vol}(D)>0$. The density of an irreducible cut-and-project set Λ along the increasing family $\{TD\}_{T>0}$, where

$$TD \stackrel{\text{def}}{=} \{t\boldsymbol{x} : \boldsymbol{x}\in D,\ t\in[0,T]\}$$

exists and is given by

$$\theta(\Lambda(\mathcal{W},\mathcal{L})) = \frac{\text{vol}(\mathcal{W})}{\text{covol}(\mathcal{L})},$$

see for example [MS14, Proposition 3.2]. In Section 2.2 we define cut-and-project sets in greater generality.

Marklof and Strömbergsson [MS15, Theorem 1] proved that, for any irreducible cut-and-project set (2), the density of $\Lambda_{\text{vis},\{\mathbf{0}\}}$ exists and satisfies

$$0 < \theta(\Lambda_{\text{vis},\{\mathbf{0}\}}) \leqslant \theta(\Lambda). \tag{3}$$

Hammarhjelm [Ham22] gave examples of cut-and-project sets $\Lambda(\mathcal{W},\mathcal{L})\subset\mathbb{R}^2$ for which the inequality in (3) is strict and computed $\theta(\Lambda_{\text{vis},\{\mathbf{0}\}})$ explicitly. The constructions in [Ham22] include the Ammann–Beenker point set and the vertex set of the Penrose tiling. Some of these examples were also considered by Sing [Sin]. These constructions are based on algebraic lattices over real quadratic fields.

To state our results, we first recall the necessary notions from algebraic number theory. Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field, where $\mathfrak{d}\geqslant 2$ is a square-free integer, and let $\mathcal{O}_K$ be its ring

of integers. We assume throughout that $\mathcal{O}_K$ is a principal ideal domain. The norm of an element $x \in \mathcal{O}_K$ is defined by

$$N(x) = x\sigma(x),$$

where σ is the nontrivial Galois automorphism of K. The *unit group* $\mathcal{O}_K^\times$ consists of the elements $x \in \mathcal{O}_K$ such that $N(x) = \pm 1$. The unique element $\lambda = \lambda_K \in \mathcal{O}_K^\times$ such that

$$\mathcal{O}_K^\times = \{\pm 1\} \times \{\lambda^i \,:\, i \in \mathbb{Z}\}, \qquad \lambda > 1,$$

is called the *fundamental unit* of K. Let $I \subseteq \mathcal{O}_K$ be a nontrivial ideal. The *norm* of I is defined by

$$N(I) \stackrel{\text{def}}{=} \#(\mathcal{O}_K/I),$$

which is always finite. Moreover, $N(I) = |N(x)|$, where I is generated by x. The Dedekind zeta function of K is then defined by

$$\zeta_{\mathcal{O}_K}(s) \stackrel{\text{def}}{=} \sum_{\substack{I \subseteq \mathcal{O}_K \\ I \neq (0)}} \frac{1}{N(I)^s}, \qquad \Re(s) > 1.$$

We refer the reader to [Mar18] for further details.

The Minkowski map

$$f : K \hookrightarrow \mathbb{R}^2, \qquad \alpha \mapsto (\alpha, \sigma(\alpha)),$$

embeds $\mathcal{O}_K$ as a lattice $f(\mathcal{O}_K)$ in $\mathbb{R}^2$. We say that K satisfies the *Hammarhjelm condition* if

$$f(\mathcal{O}_K) \cap ((1, \lambda) \times [-1, 1]) = \varnothing. \tag{4}$$

For $2 \leqslant \mathfrak{d} \leqslant 100$, the Hammarhjelm condition is satisfied only for

$$\mathfrak{d} = 2, 5, 13, 29, 53;$$

see [GKW25; Ham22]. Recently, Rudnick [Rud26] completed the classification by proving that the real quadratic fields satisfying the Hammarhjelm condition are precisely those with

$$\mathfrak{d} = 2, 5, 13, 29, 53, 173, 293.$$

In Lemma 3.2, we strengthen the Hammarhjelm condition (4) by proving that, for each of these fields,

$$f(\mathcal{O}_K) \cap \left((1, \lambda) \times \left[-\left(1 + \frac{1}{\lambda}\right), \left(1 + \frac{1}{\lambda}\right)\right]\right) = \varnothing,$$

which is crucial for the geometric condition on the finite set $S \subset \Lambda(\mathcal{W}, \mathcal{L})$ in Theorem 1.1.

By abuse of notation, for $\boldsymbol{x} = (x_1, \ldots, x_d) \in K^d$ we write

$$\sigma(\boldsymbol{x}) \stackrel{\text{def}}{=} (\sigma(x_1), \ldots, \sigma(x_d)).$$

For every $d \in \mathbb{N}$,

$$\mathcal{L} \stackrel{\text{def}}{=} \{(\boldsymbol{x}, \sigma(\boldsymbol{x})) : \boldsymbol{x} \in \mathcal{O}_K^d\} \tag{5}$$

is a lattice in $\mathbb{R}^{2d}$. Let $d \geqslant 2$, let $\mathcal{L}$ be the lattice in (5), and $\mathcal{W} \subset \mathbb{R}^d$ be a bounded, convex, centrally symmetric window with nonempty interior. Suppose that the real quadratic field K satisfies the Hammarhjelm condition. For the associated cut-and-project set $\Lambda = \Lambda(\mathcal{W}, \mathcal{L})$, the density of the set of visible points from the origin, along the family $\{TD\}_{T>0}$ exists, and is given by

$$\theta(\Lambda_{\text{vis},\{\mathbf{0}\}}) = \left(1 - \frac{1}{\lambda^d}\right) \frac{\theta(\Lambda)}{\zeta_{\mathcal{O}_K}(d)}, \tag{6}$$

where D is a Jordan measurable set with $\mathrm{vol}(D) > 0$, see [Ham22, Theorems 4.9, 4.12, 4.15]. An effective rate for the density is given in [GKW25, Theorem 5.1] when D is a bounded convex set with $\mathrm{vol}(D) > 0$.

According to [GKW25, Theorem 1.1], $\Lambda_{\mathrm{vis},\{\mathbf{0}\}}$ contains arbitrarily large holes. In fact, it follows from the proof of [KO26, Theorem 1.3], together with [HuRi15, Proposition 2.12], that these holes repeat. Adjoining any zero density set to $\Lambda_{\mathrm{vis},\{\mathbf{0}\}}$ still does not produce a Delone set.

Let $S \subset \Lambda$ be a finite set and let $\mathcal{P}_{\mathcal{O}_K}$ be the set of nonzero prime ideals of $\mathcal{O}_K$. For each $\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}$, denote by

$$\pi_{\mathfrak{p}} : \mathcal{O}_K^d \to \left(\mathcal{O}_K/\mathfrak{p}\right)^d$$

the quotient map, and define

$$s(\mathfrak{p}) \stackrel{\text{def}}{=} \#(\pi_{\mathfrak{p}}(S)).$$

For $\boldsymbol{s} \in \Lambda$, define

$$\mathcal{W}_{\boldsymbol{s}} \stackrel{\text{def}}{=} \frac{1}{\lambda}\big(\mathcal{W} + (\lambda+1)\sigma(\boldsymbol{s})\big).$$

For a nonempty, finite set $I \subset \Lambda$, denote

$$\mathcal{W}_I \stackrel{\text{def}}{=} \bigcap_{\boldsymbol{s}\in I} \mathcal{W}_{\boldsymbol{s}}.$$

If $\mathcal{W}$ is convex, then each $\mathcal{W}_I$ is convex. Furthermore, if $\mathcal{W}$ is bounded, centrally symmetric, has nonempty interior, and for every $\boldsymbol{s} \in S$ we have the geometric condition

$$-(\mathcal{W} - \sigma(\boldsymbol{s})) \subset \left(1 + \frac{1}{\lambda}\right)(\mathcal{W} - \sigma(\boldsymbol{s})), \tag{7}$$

then (see Lemma 4.1)

$$\mathcal{W}_{\boldsymbol{s}} \subset \mathcal{W}.$$

In the following theorem, we prove that the density of the set of simultaneously visible points from a finite set S along $\{TD\}_{T>0}$ exists and derive an explicit formula for it, assuming that each $\boldsymbol{s} \in S$ satisfies the condition (7).

Theorem 1.1. *Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field with ring of integers $\mathcal{O}_K$ satisfying the Hammarhjelm condition. Let $d \geqslant 2$, let $\mathcal{L}$ be the lattice defined in (5), and let $\mathcal{W} \subset \mathbb{R}^d$ be a bounded, convex, centrally symmetric window with nonempty interior. Let $\Lambda = \Lambda(\mathcal{W}, \mathcal{L})$ be the associated cut-and-project set. Suppose that $S \subset \Lambda$ is finite and that (7) holds for every $\boldsymbol{s} \in S$. Then the density of $\Lambda_{\mathrm{vis},S}$ along $\{TD\}_{T>0}$ exists and is given by*

$$\theta(\Lambda_{\mathrm{vis},S}) = \mathrm{dens}(\mathcal{L}) \left(\mathrm{vol}(\mathcal{W}) - \sum_{\varnothing \neq I \subseteq S} (-1)^{\#(I)+1} \mathrm{vol}(\mathcal{W}_I)\right) \prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right).$$

Here $D \subset \mathbb{R}^d$ is a bounded convex set with $\mathrm{vol}(D) > 0$.

It has been shown in [BMP00] that, for $k \geqslant 2$, the set $\mathbb{Z}^k_{\mathrm{vis},\{\mathbf{0}\}}$ is a weak model set of maximal density; see Section 2.3 for the definition. Moreover, it has been shown in [KO26, Theorem 1.1] that if $\mathbb{Z}^k_{\mathrm{vis},S}$ has positive density along $[-L, L]^k$, then $\mathbb{Z}^k_{\mathrm{vis},S}$ is a weak model set of maximal density. Furthermore, it follows from the proof of [KO26, Theorem 1.3] that the set $\Lambda_{\mathrm{vis},\{\mathbf{0}\}}$ is also a weak model set of maximal density along the sequence $\{TD\}_{T>0}$. The following corollary extends this result to the set $\Lambda_{\mathrm{vis},S}$.

Corollary 1.2. *Under the hypotheses of Theorem 1.1, suppose that the density $\theta(\Lambda_{\mathrm{vis},S})$ is positive along $\{TD\}_{T>0}$. Then the set $\Lambda_{\mathrm{vis},S}$ is a weak model set of maximal density.*

Denote by

$$\mathscr{L}_n \stackrel{\text{def}}{=} \mathrm{SL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{Z}), \qquad \mathscr{A}_n \stackrel{\text{def}}{=} \mathrm{ASL}_n(\mathbb{R})/\mathrm{ASL}_n(\mathbb{Z}),$$

the quotient spaces that parameterize covolume-one lattices and affine lattices in $\mathbb{R}^n$, respectively. Denote by $\mathrm{m}_{\mathscr{L}_n}$ and $\mathrm{m}_{\mathscr{A}_n}$ the probability measures on $\mathscr{L}_n$ and $\mathscr{A}_n$ induced by Haar measures on $\mathrm{SL}_n(\mathbb{R})$ and $\mathrm{ASL}_n(\mathbb{R})$, respectively. Fix the decomposition in (1) and a bounded measurable

window $\mathcal{W} \subset \mathbb{R}^m$ with nonempty interior and whose boundary has Lebesgue measure zero. The map

$$\Psi(\mathcal{L}) \stackrel{\text{def}}{=} \Lambda(\mathcal{W}, \mathcal{L})$$

associates a cut-and-project set with each lattice or affine lattice. We denote by

$$\mu \stackrel{\text{def}}{=} \Psi_* \mathrm{m}_{\mathscr{L}_n}, \qquad \overline{\mu} \stackrel{\text{def}}{=} \Psi_* \mathrm{m}_{\mathscr{A}_n}$$

the resulting push-forward measures on the space of closed subsets of $\mathbb{R}^d$. These are, respectively, linear and affine Ratner–Marklof–Strömbergsson measures first considered in [MS15] and then classified in [RSW24]. We review the general definition of RMS measures in Section 2.4.

Let $\mathcal{L} \subset \mathbb{R}^n$ be a lattice, and let $\tilde{S} \subset \mathcal{L}$. For each $p \in \mathcal{P}$, let

$$\pi_{\mathcal{L},p} : \mathcal{L} \to \mathcal{L}/p\mathcal{L}\,,$$

be the natural projection. We write

$$s_{\mathcal{L}}(p) \stackrel{\text{def}}{=} \#\Big(\pi_{\mathcal{L},p}(\tilde{S})\Big)\,.$$

Theorem 1.3. *Let $n = d + m$, where $d \geqslant 2$ and $m \geqslant 1$, and let $\mathcal{W} \subset \mathbb{R}^m$ be a bounded, convex window with nonempty interior and boundary of Lebesgue measure zero. Let $\Lambda = \Lambda(\mathcal{W}, \mathcal{L})$ be an irreducible cut-and-project set. Suppose that*

$$\dim\left(\operatorname{span}_{\mathbb{R}}\{\boldsymbol{v}_1, \boldsymbol{v}_2\} \cap (\{\mathbf{0}\} \times \mathbb{R}^m)\right) = 0 \tag{8}$$

for every pair of linearly independent vectors $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathcal{L}$. Then, for every finite set $S \subset \Lambda$, the density of $\Lambda_{\mathrm{vis},S}$ along $\{TD\}_{T>0}$ exists and is given by

$$\theta(\Lambda_{\mathrm{vis},S}) = \frac{\operatorname{vol}(\mathcal{W})}{\operatorname{covol}(\mathcal{L})} \prod_{p \in \mathcal{P}} \left(1 - \frac{s_{\mathcal{L}}(p)}{p^n}\right),$$

where $D \subset \mathbb{R}^d$ is bounded and convex with positive volume, $\tilde{S} \stackrel{\text{def}}{=} (\pi_{\mathrm{phys}}|_{\mathcal{L}})^{-1}(S)$. Moreover, condition (8) holds for μ-almost every irreducible cut-and-project set. Consequently, the formula above holds for μ-almost every $\Lambda(\mathcal{W}, \mathcal{L})$ and every finite $S \subset \Lambda$.

Remark 1.4. In [GKW25, Theorem 3.4], for $S = \{\mathbf{0}\}$, the density formula in the preceding theorem was obtained for μ-almost every cut-and-project set together with an error term.

We prove Theorem 1.3 in Section 6.

Proposition 1.5. *Let $d \geqslant 2$, let $\mathcal{W} \subset \mathbb{R}^m$ be a bounded window with nonempty interior and boundary of Lebesgue measure zero, and let $S \subset \mathbb{R}^d$ be countable. Then, for $\overline{\mu}$-almost every cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$,*

$$\Lambda(\mathcal{W}, \mathcal{L}) \backslash S = \Lambda(\mathcal{W}, \mathcal{L})_{\mathrm{vis},S}.$$

Therefore,

$$\theta\left(\Lambda(\mathcal{W}, \mathcal{L})_{\mathrm{vis},S}\right) = \frac{\operatorname{vol}(\mathcal{W})}{\operatorname{covol}(\mathcal{L})}.$$

1.3. **Remarks.**

(1) During the preparation of this manuscript, the authors became aware of recent work by H. Shamir [Sha] that generalizes the density formula (6). In particular, Shamir avoids the Hammarhjelm condition and proves the result for every real quadratic field $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$, where $\mathfrak{d} \geqslant 2$ is a square-free integer.

(2) It follows from [BHS17, Theorem 7] and Corollary 1.2 that the diffraction spectrum of $\Lambda_{\mathrm{vis},S}$ is pure point and translation bounded. We refer the reader to [Sch00; Hof95; KO26] and the references therein for details on the diffraction.

(3) Under the hypotheses of Theorem 1.3, if $\theta(\Lambda_{\mathrm{vis},S}) > 0$, then it follows from the proof of Theorem 1.3 that the set $\Lambda_{\mathrm{vis},S}$ is a weak model set of maximal density along $\{TD\}_{T>0}$.

(4) We stated Theorem 1.3 and Proposition 1.5 for the measures μ and $\overline{\mu}$, respectively. However, extending these results with respect to arbitrary RMS measures remains an open problem.

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2. Preliminaries

2.1. **Averaging sequences.** Let G be a σ-compact locally compact abelian group equipped with a Haar measure θ_G. An *averaging sequence* is a family $\{A_T\}_{T\in\mathcal{I}}$ of precompact Borel subsets of G, indexed by an unbounded set $\mathcal{I}\subset(0,\infty)$, such that

(1) for $T<T'$, $A_T\subseteq A_{T'}$,
(2) $\lim_{T\to\infty}\theta_G(A_T)=\theta_G(G)$, and
(3) $\bigcup_{T\in\mathcal{I}}A_T=G$ up to measure zero.

An averaging sequence is a *van Hove sequence* if $\theta_G(A_T)>0$ for all sufficiently large T, and for every compact set $K\subseteq G$, the K-boundary is asymptotically negligible, that is,

$$\lim_{T\to\infty}\frac{\theta_G\big(\partial^K A_T\big)}{\theta_G(A_T)}=0,$$

where

$$\partial^K A_T \stackrel{\text{def}}{=} \big((K+A_T)\backslash A_T^{\mathrm{o}}\big)\cup\big((-K+\overline{A_T^{\mathsf{c}}})\cap A_T\big)$$

is the K-boundary of A_T. Here A_T^{c}, ${A_T}^{\mathrm{o}}$ and $\overline{A_T}$ denote the complement, interior and closure of $A_T\subset G$, respectively, see [Moo02, p. 125], [Sch98, p. 145], and [HuRi15, p.746]. The existence of van Hove sequences in σ-compact LCAGs is shown in [Sch98]. In the Euclidean setting, namely $G=\mathbb{R}^d$, standard examples of van Hove sequences include increasing sequences of Euclidean balls and cubes. More generally, truncated cones $TA=\{t\boldsymbol{x}:\boldsymbol{x}\in A,\ t\in[0,T]\}$ of a bounded Jordan set $A\subset\mathbb{R}^d$ with nonempty interior:

Lemma 2.1. *Let $A\subset\mathbb{R}^d$ be bounded measurable set with nonempty interior, and suppose that ∂A has Lebesgue measure zero. Additionally, suppose that $\mathbf{0}\in A^{\mathrm{o}}$. Then the family $\{TA\}_{T>0}$ forms a van Hove sequence.*

We omit the proof of this lemma, since it follows directly from the definitions. We shall use the fact that every bounded convex subset of $\mathbb{R}^d$ has boundary of Lebesgue measure zero and therefore Jordan measurable [Sza97].

The upper and lower densities of $\Lambda\subset G$ with respect to an averaging sequence $\{A_T\}_{T\in\mathcal{I}}$ are defined by

$$\overline{\theta}(\Lambda)\stackrel{\text{def}}{=}\limsup_{T\to\infty}\frac{\#(\Lambda\cap A_T)}{\theta_G(A_T))}\qquad\text{and}\qquad\underline{\theta}(\Lambda)\stackrel{\text{def}}{=}\liminf_{T\to\infty}\frac{\#(\Lambda\cap A_T)}{\theta_G(A_T))},$$

respectively.

2.2. **General cut-and-project sets.** Let G and H be σ-compact, locally compact abelian groups (LCAGs) equipped with Haar measures θ_G and θ_H, respectively. Throughout, we assume that H is compactly generated. By the Structure Theorem for compactly generated locally compact abelian groups, H is topologically isomorphic to $\mathbb{R}^a\times\mathbb{Z}^b\times\mathbb{K}$, for some integers $a,b\geqslant 0$ and some compact abelian group $\mathbb{K}$, see [HeRo79, Theorem 9.8].

We refer to G as the *physical space* and to H as the *internal space*. Let $\mathcal{L}\subset G\times H$ be a lattice, and let $\mathcal{W}\subset H$ be a measurable window. Let

$$\pi_{\mathrm{phys}}:G\times H\to G,\qquad \pi_{\mathrm{int}}:G\times H\to H$$

denote the natural projections. We assume that the restriction $\pi_{\mathrm{phys}}|_{\mathcal{L}} : \mathcal{L} \to G$ is injective and that the image $\pi_{\mathrm{int}}(\mathcal{L})$ is dense in H. The associated cut-and-project set is

$$\Lambda(\mathcal{W}, \mathcal{L}) \stackrel{\mathrm{def}}{=} \{\pi_{\mathrm{phys}}(\boldsymbol{z}) : \boldsymbol{z} \in \mathcal{L},\ \pi_{\mathrm{int}}(\boldsymbol{z}) \in \mathcal{W}\}.$$

The cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ is called a *model set* if the window $\mathcal{W}$ is relatively compact and has nonempty interior, see [BHS17] and the references therein. If, in addition, $\mathcal{W} = \overline{\mathcal{W}^{\mathrm{o}}}$ and $\mathcal{W}$ is compact, then the window is called *proper*. If $\mathcal{W}$ is a proper window, and satisfies

$$\theta_H(\partial\mathcal{W}) = 0,$$

then the window $\mathcal{W}$ is called *regular* and the cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ is called *irreducible*.

The density of an irreducible cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ along a van Hove sequence $\{A_n\}_{n\in\mathbb{N}}$ exists and is given by

$$\theta(\Lambda(\mathcal{W}, \mathcal{L})) = \lim_{n\to\infty} \frac{\#(\Lambda(\mathcal{W}, \mathcal{L}) \cap A_n)}{\theta_G(A_n)} = \frac{\theta_H(\mathcal{W})}{\mathrm{covol}(\mathcal{L})}, \tag{9}$$

see [Moo02, Theorem 1], [Sch98], [HuRi15, Lemma 3.2, Remark 3.3], [MS14, Proposition 3.2], and the references therein. Moreover, the density of irreducible cut-and-project sets is independent of the van Hove sequence, see [HuRi15, Remark 3.3(i)]. Irreducible cut-and-project sets are Delone sets, see [HuRi15, p. 744].

Recall that a set $\mathcal{D} \subset G$ is *uniformly discrete* if there exists an open neighborhood U of the identity in G such that any of its translates $x + U$, with $x \in G$, contains at most one point of $\mathcal{D}$. The set $\mathcal{D} \subset G$ is *relatively dense* if there exists a compact set $K \subset G$ such that $\mathcal{D} + K = G$. A subset $\mathcal{D} \subset G$ is called a *Delone set* if it is both uniformly discrete and relatively dense.

In Theorem 1.1, the window $\mathcal{W} \subset \mathbb{R}^m$ is assumed to be bounded and convex, to have nonempty interior, and to be centrally symmetric about the origin. The latter condition means that

$$\mathcal{W} = -\mathcal{W}.$$

2.3. **Weak model sets of maximal density.** Let G and H be as in Section 2.2. A cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ is called a *weak model set* if the window $\mathcal{W}$ is relatively compact and satisfies

$$\theta_H(\overline{\mathcal{W}}) > 0.$$

Unlike model sets, we do not require the window to have a nonempty interior. Weak model sets are uniformly discrete (see [HuRi15, Proposition 2.4]). Furthermore, if $\mathcal{W}$ is nowhere dense, then the cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ contains arbitrarily large holes and has the hole-repetitive property, see [HuRi15, Proposition 2.12] for the definition and proof of this. Consequently, it is not relatively dense.

We say that $\Lambda(\mathcal{W}, \mathcal{L})$ is a *weak model set of maximal density* relative to a given averaging sequence $\mathcal{A}$ if $\Lambda(\mathcal{W}, \mathcal{L})$ is a weak model set, and if the density of $\Lambda(\mathcal{W}, \mathcal{L})$ relative to $\mathcal{A}$ exists and is given by

$$\theta\big(\Lambda(\mathcal{W}, \mathcal{L})\big) = \frac{\theta_H(\overline{\mathcal{W}})}{\mathrm{covol}(\mathcal{L})}.$$

We refer the reader to [BHS17] and the reference therein for further details.

2.4. **Ratner–Marklof–Strömbergsson measures.** Let $\mathrm{SL}_n(\mathbb{R})$ be the group of linear transformations of $\mathbb{R}^n$ that are volume and orientation preserving. Let $\mathrm{ASL}_n(\mathbb{R}) = \mathrm{SL}_n(\mathbb{R}) \ltimes \mathbb{R}^n$ be the group of affine transformations on $\mathbb{R}^n$ that are volume and orientation preserving. Consider the homogeneous spaces

$$\mathscr{L}_n \stackrel{\mathrm{def}}{=} \mathrm{SL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{Z}) \quad \text{and} \quad \mathscr{A}_n \stackrel{\mathrm{def}}{=} \mathrm{ASL}_n(\mathbb{R})/\mathrm{ASL}_n(\mathbb{Z})$$

both equipped with their quotient topologies. These spaces parameterize, respectively, covolume-one lattices and affine lattices in $\mathbb{R}^n$. We write $\mathrm{m}_{\mathscr{L}_n}$ and $\mathrm{m}_{\mathscr{A}_n}$ for the corresponding Haar-Siegel

measures. These are the unique probability measures invariant under the actions of $\mathrm{SL}_n(\mathbb{R})$ and $\mathrm{ASL}_n(\mathbb{R})$, respectively.

Let $\mathscr{C}(\mathbb{R}^n)$ be the collection of closed subsets of $\mathbb{R}^n$. Equipped with the *Chabauty-Fell topology*, $\mathscr{C}(\mathbb{R}^n)$ is compact, see [RSW24, § 2.2] and the references therein. After identifying $\mathscr{L}_n$ and $\mathscr{A}_n$ as subsets of $\mathscr{C}(\mathbb{R}^n)$, the restriction of the Chabauty-Fell topology to these, coincides with their quotient topology. Therefore, lattices and affine lattices satisfy natural inclusions

$$\mathscr{L}_n \subset \mathscr{A}_n \subset \mathscr{C}(\mathbb{R}^n)$$

as topological subspaces.

The natural actions of $\mathrm{ASL}_d(\mathbb{R})$ and $\mathrm{SL}_d(\mathbb{R})$ on $\mathbb{R}^d$ induce corresponding actions on $\mathscr{C}(\mathbb{R}^d)$. A probability measure on $\mathscr{C}(\mathbb{R}^d)$ is called a *Ratner-Marklof-Strömbergsson linear measure*, or an *RMS linear measure* for short, if it is invariant and ergodic with respect to the $\mathrm{SL}_d(\mathbb{R})$-action and gives full measure to the set of irreducible cut-and-project sets.

Identifying the group $\mathrm{SL}_d(\mathbb{R})$ as a subgroup of $\mathrm{ASL}_d(\mathbb{R})$ in the canonical way, an RMS linear measure does not need to be invariant under the $\mathrm{ASL}_d(\mathbb{R})$ action. When this additional invariance property holds, the measure is called an *RMS affine measure*.

Fix the decomposition in (1), the associated projections π_{phys} and π_{int}, a bounded measurable window $\mathcal{W} \subset \mathbb{R}^m$ with nonempty interior and boundary of Lebesgue measure zero. Define

$$\Psi_{\mathcal{W}} : \mathscr{A}_n \longrightarrow \mathscr{C}(\mathbb{R}^d), \qquad \Psi_{\mathcal{W}}(\mathcal{L}) \stackrel{\text{def}}{=} \Lambda(\mathcal{W}, \mathcal{L}).$$

The map $\Psi_{\mathcal{W}}$ is Borel measurable ([RSW24, Proposition 2.1]), and hence we may define the probability measures

$$\mu \stackrel{\text{def}}{=} (\Psi_{\mathcal{W}})_* \mathrm{m}_{\mathscr{L}_n}, \qquad \overline{\mu} \stackrel{\text{def}}{=} (\Psi_{\mathcal{W}})_* \mathrm{m}_{\mathscr{A}_n}$$

on $\mathscr{C}(\mathbb{R}^d)$. It was proved that the measures μ and $\overline{\mu}$ are RMS linear and affine measures, respectively, see [RSW24, Theorem 4.1]. We refer the reader to [MS14; RSW24] for further details on RMS measures.

For a lattice $\mathcal{L} \subset \mathbb{R}^d \oplus \mathbb{R}^m$ and a pair of linearly independent vectors $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathcal{L}$, define

$$\dim_{\mathcal{L}}(\boldsymbol{v}_1, \boldsymbol{v}_2) \stackrel{\text{def}}{=} \dim\left(\mathrm{span}_{\mathbb{R}}\{\boldsymbol{v}_1, \boldsymbol{v}_2\} \cap (\{\mathbf{0}\} \times \mathbb{R}^m)\right). \tag{10}$$

This quantity can be calculated for any given lattice. Moreover, for almost every $g \in \mathrm{GL}_n(\mathbb{R})$, the lattice $\mathcal{L} = g\mathbb{Z}^n$ and every pair of linearly independent vectors satisfies $\dim_{\mathcal{L}}(v_1, v_2) = 0$, see the proof of [GKW25, Proposition 3.3]. Thus, any statement proved under condition (8) applies to μ-almost every irreducible cut-and-project set.

2.5. **Visibility from a point in a cut-and-project set.** Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field with ring of integers $\mathcal{O}_K$ satisfying the Hammarhjelm condition. For two distinct points $\boldsymbol{x} = (x_1, \ldots, x_k)$ and $\boldsymbol{y} = (y_1, \ldots, y_k)$ in $\mathcal{O}_K^k$, we define

$$\gcd(\boldsymbol{x} - \boldsymbol{y}) = \gcd\big((x_1 - y_1), \ldots, (x_k - y_k)\big),$$

where $(x_i - y_i)$ denotes the ideal generated by $x_i - y_i$. Since $\mathcal{O}_K$ is a principal ideal domain, every such ideal is principal. By abuse of notation, we write $\gcd(\boldsymbol{x} - \boldsymbol{y})$ for a generator of this ideal. In particular, if

$$\gcd(\boldsymbol{x} - \boldsymbol{y}) = \mathcal{O}_K,$$

we simply write

$$\gcd(\boldsymbol{x} - \boldsymbol{y}) = 1.$$

We also define

$$\gcd(\mathbf{0}) \stackrel{\text{def}}{=} 0.$$

Let $\mathcal{W} \subset \mathbb{R}^d$ be a bounded, convex, centrally symmetric window with nonempty interior, and let $\mathcal{L}$ be the lattice defined in (5). By [GKW25, Lemma 5.6],

$$\Lambda_{\mathrm{vis},\{\mathbf{0}\}}(\mathcal{W},\mathcal{L}) = \left\{ \boldsymbol{x} \in \Lambda(\mathcal{W},\mathcal{L})\backslash\{\mathbf{0}\} : \gcd(\boldsymbol{x}) = 1,\ \sigma(\boldsymbol{x}) \notin \frac{1}{\lambda}\mathcal{W} \right\}.$$

The following lemma extends this characterization to visibility from an arbitrary point $\boldsymbol{s} \in \Lambda(\mathcal{W},\mathcal{L})$ satisfying a suitable geometric condition.

Lemma 2.2. *Under the hypotheses of Theorem 1.1, let $\boldsymbol{s} \in \Lambda$ satisfy the geometric condition (7). Then the set of points visible from $\boldsymbol{s}$ is given by*

$$\Lambda_{\mathrm{vis},\{\boldsymbol{s}\}} = \left\{ \boldsymbol{x} \in \Lambda : \gcd(\boldsymbol{x}-\boldsymbol{s}) = 1,\ \sigma\left(\frac{\boldsymbol{x}-\boldsymbol{s}}{\lambda}\right) \notin \mathcal{W} - \sigma(\boldsymbol{s}) \right\}. \tag{11}$$

3. Proof of Lemma 2.2

Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field satisfying the Hammarhjelm condition, and let $\lambda > 1$ be its fundamental unit. Rudnick [Rud26] shows that

$$\mathfrak{d} \in \{2, 5, 13, 29, 53, 173, 293\}.$$

Observe that if

$$\mathfrak{d} \in \{13, 29, 53, 173, 293\},$$

then

$$\mathfrak{d} = x^2 + 4, \qquad x \in \{3, 5, 7, 13, 17\}.$$

Writing $x = 2m + 1$, we have

$$\mathcal{O}_K = \mathbb{Z}[\omega], \qquad \omega = \frac{1+\sqrt{\mathfrak{d}}}{2},$$

and

$$\lambda = \frac{x+\sqrt{\mathfrak{d}}}{2} = m + \omega.$$

In particular,

$$N(\lambda) = -1, \qquad \sigma(\lambda) = -\frac{1}{\lambda},$$

and

$$\frac{1}{\lambda} = \frac{\sqrt{\mathfrak{d}}-x}{2}, \qquad 1 + \frac{1}{\lambda} = \frac{\sqrt{\mathfrak{d}}-x+2}{2}.$$

Lemma 3.1. *Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field and let λ be its fundamental unit. Then*

$$1 + \frac{1}{\lambda} \leqslant \lambda. \tag{12}$$

Equality occurs only for $K = \mathbb{Q}(\sqrt{5})$.

Proof. It can easily be seen that the condition (12) is equivalent to

$$\lambda \geqslant \frac{1+\sqrt{5}}{2}.$$

Case 1: $\mathfrak{d} = 2, 3 \bmod 4$.

In this case, the ring of integers is $\mathbb{Z}[\sqrt{\mathfrak{d}}]$. We claim that the fundamental unit $\lambda = m + n\sqrt{\mathfrak{d}}$ for some $n, m \in \mathbb{N}$. In fact, if $n = 0$, then $N(\lambda) = m^2 = \pm 1$, so $m = \pm 1$, which is contradiction of $\lambda > 1$. Now if $n < 0$, then $\sigma(\lambda) = m - n\sqrt{\mathfrak{d}} > m + n\sqrt{\mathfrak{d}} = \lambda > 1$, which contradicts $|N(\lambda)| = \lambda\sigma(\lambda) = 1$.

Now, if $m = 0$ then we have $N(\lambda) = -n^2\mathfrak{d} \neq \pm 1$. So we must have $m \neq 0$. Suppose that $m < 0$, then $\sigma(\lambda) = m - n\sqrt{\mathfrak{d}} < 0$. Therefore

$$\lambda = \frac{1}{|\sigma(\lambda)|} = \frac{1}{n\sqrt{\mathfrak{d}} - m} \leqslant \frac{1}{n\sqrt{\mathfrak{d}}} < 1,$$

which contradicts $\lambda > 1$.

Since $n, m \geqslant 1$, we have

$$\lambda = m + n\sqrt{\mathfrak{d}} \geqslant 1 + \sqrt{\mathfrak{d}} \geqslant 1 + \sqrt{2} > \frac{1+\sqrt{5}}{2}$$

Case 2: $\mathfrak{d} = 1 \bmod 4$.

In this case the ring of integers is $\mathbb{Z}[\omega]$, where $\omega = (1+\sqrt{\mathfrak{d}})/2$. By [Rud26, Lemma 2.1], we have $\lambda = m + n\omega$ with $m \geqslant 0$, $n \geqslant 1$. Thus

$$\lambda \geqslant 0 + \omega = (1+\sqrt{\mathfrak{d}})/2 \geqslant \frac{1+\sqrt{5}}{2},$$

because $\mathfrak{d} > 1$, and equality if and only if $\mathfrak{d} = 5$. □

Denote

$$\mathbb{P} \stackrel{\text{def}}{=} \{\pi \in \mathcal{O}_K : \pi \text{ is prime and } 1 < \pi < \lambda\}.$$

The set $\mathbb{P}$ plays the role of a fundamental domain for the action of the unit group on prime elements of $\mathcal{O}_K$: after multiplying by a unit, and possibly by -1, every prime element has a representative in $\mathbb{P}$.

It follows from [Ham22, Lemma 4.7] that, if $\mathfrak{d} = 2$, then

$$|\sigma(\pi)| \geqslant 1 + \frac{1}{\lambda} \qquad \text{for all } \pi \in \mathbb{P},$$

and if $\mathfrak{d} = 5$, then [Ham22, Lemma 4.13] gives the stronger bound

$$|\sigma(\pi)| \geqslant 2\lambda \qquad \text{for all } \pi \in \mathbb{P}.$$

Lemma 3.2. *Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field satisfying the Hammarhjelm condition. Then, for every $\pi \in \mathbb{P}$,*

$$|\sigma(\pi)| \geqslant 1 + \frac{1}{\lambda}.$$

Moreover, for $\mathfrak{d} \in \{2, 13, 29, 53, 173, 293\}$, this bound is sharp.

Proof. By [Ham22, Lemmas 4.7 and 4.13], the desired lower bound holds for $\mathfrak{d} = 2, 5$. Moreover, for $\mathfrak{d} = 2$, the prime element $\pi = \sqrt{2}$ satisfies

$$1 < \sqrt{2} < \lambda, \qquad |\sigma(\sqrt{2})| = \sqrt{2} = 1 + \frac{1}{\lambda},$$

so the bound is sharp in this case. It remains to treat

$$\mathfrak{d} \in \{13, 29, 53, 173, 293\}.$$

As discussed at the beginning of Section 3, there is $x \in \{3, 5, 7, 13, 17\}$ such that

$$\mathfrak{d} = x^2 + 4.$$

Writing $x = 2m + 1$, we have

$$\mathcal{O}_K = \mathbb{Z}[\omega], \qquad \omega = \frac{1+\sqrt{\mathfrak{d}}}{2}, \qquad \lambda = m + \omega.$$

Also,

$$\sigma(\lambda) = -\frac{1}{\lambda}, \qquad \sigma(\omega) = -m - \frac{1}{\lambda}.$$

Let $\pi \in \mathbb{P}$. There exist unique integers $a, b \in \mathbb{Z}$ such that

$$\pi = a + b\omega.$$

Since $\omega - \sigma(\omega) = \sqrt{\mathfrak{d}}$, we have

$$\sigma(\pi) = \pi - b\sqrt{\mathfrak{d}}. \tag{13}$$

We divide the proof into cases according to b.

- If $b \leqslant -1$, then by (13),
$$\sigma(\pi) = \pi + |b|\sqrt{\mathfrak{d}} > 1 + \sqrt{\mathfrak{d}} > \sqrt{\mathfrak{d}}.$$
Since $1 + \lambda^{-1} < 2 < \sqrt{\mathfrak{d}}$, it follows that
$$|\sigma(\pi)| > 1 + \frac{1}{\lambda}.$$
- If $b = 0$, then $\pi = a \in \mathbb{Z}$. Since $\pi > 1$, we have $a \geqslant 2$, and hence
$$|\sigma(\pi)| = a \geqslant 2 > 1 + \frac{1}{\lambda}.$$
- If $b \geqslant 2$, then by (13),
$$\sigma(\pi) = \pi - b\sqrt{\mathfrak{d}} \leqslant \pi - 2\sqrt{\mathfrak{d}}.$$
Since $\mathfrak{d} = x^2 + 4$, we have $x < \sqrt{\mathfrak{d}}$, and therefore
$$\lambda = \frac{x + \sqrt{\mathfrak{d}}}{2} < \sqrt{\mathfrak{d}}.$$
As $\pi < \lambda$, we get
$$\sigma(\pi) < \sqrt{\mathfrak{d}} - 2\sqrt{\mathfrak{d}} = -\sqrt{\mathfrak{d}}.$$
Since $\sqrt{\mathfrak{d}} > 2 > 1 + \lambda^{-1}$, it follows that
$$|\sigma(\pi)| > 1 + \frac{1}{\lambda}.$$
- If $b = 1$, then $\pi = a + \omega$. The inequality $\pi < \lambda = m + \omega$ gives $a \leqslant m - 1$. Hence
$$\sigma(\pi) = a + \sigma(\omega) = a - m - \frac{1}{\lambda} \leqslant -1 - \frac{1}{\lambda}.$$
Thus
$$|\sigma(\pi)| \geqslant 1 + \frac{1}{\lambda}.$$

It remains to prove sharpness for

$$\mathfrak{d} \in \{13, 29, 53, 173, 293\}.$$

Set

$$\pi_0 = \lambda - 1.$$

Since $\lambda > x \geqslant 3$, we have

$$1 < \pi_0 < \lambda.$$

Moreover,

$$|\sigma(\pi_0)| = |\sigma(\lambda) - 1| = \left|-\frac{1}{\lambda} - 1\right| = 1 + \frac{1}{\lambda}.$$

Finally,

$$N(\lambda - 1) = -x.$$

Since $x \in \{3, 5, 7, 13, 17\}$ is a rational prime, the ideal $(\lambda - 1)$ has prime norm. Hence $(\lambda - 1)$ is a prime ideal, so $\lambda - 1$ is a prime element. Therefore $\pi_0 \in \mathbb{P}$, and the bound is sharp. □

Proof of Lemma 2.2. We first prove the inclusion $\subseteq$ in (11). Let $\boldsymbol{x} \in \Lambda_{\mathrm{vis},\{\boldsymbol{s}\}}$. We show that

$$\gcd(\boldsymbol{x}-\boldsymbol{s}) = 1 \qquad \text{and} \qquad \sigma\left(\frac{\boldsymbol{x}-\boldsymbol{s}}{\lambda}\right) \notin \mathcal{W} - \sigma(\boldsymbol{s}).$$

Suppose first that

$$\sigma\left(\frac{\boldsymbol{x}-\boldsymbol{s}}{\lambda}\right) \in \mathcal{W} - \sigma(\boldsymbol{s}).$$

Then

$$\boldsymbol{y}' = \boldsymbol{s} + \frac{\boldsymbol{x}-\boldsymbol{s}}{\lambda}$$

belongs to Λ, and since $\lambda > 1$, the point $\boldsymbol{y}'$ lies in the interior of the line segment joining $\boldsymbol{s}$ and $\boldsymbol{x}$. This contradicts the visibility of $\boldsymbol{x}$ from $\boldsymbol{s}$.

Next, suppose that $\gcd(\boldsymbol{x}-\boldsymbol{s}) \neq 1$. Then there exists a prime element $\pi \in \mathcal{O}_K$ dividing $\boldsymbol{x}-\boldsymbol{s}$. After multiplying π by a unit, and possibly by -1, we may assume that $\pi \in \mathbb{P}$. Set

$$\boldsymbol{y} = \boldsymbol{s} + \frac{\boldsymbol{x}-\boldsymbol{s}}{\pi}.$$

Then $\boldsymbol{y} \in \mathcal{O}_K^d$, and since $1 < \pi < \lambda$, the point $\boldsymbol{y}$ lies in the interior of the line segment joining $\boldsymbol{s}$ and $\boldsymbol{x}$. It remains to show that $\boldsymbol{y} \in \Lambda(\mathcal{W}, \mathcal{L})$. We have

$$\sigma(\boldsymbol{y}) = \sigma(\boldsymbol{s}) + \sigma(\pi)^{-1}\sigma(\boldsymbol{x}-\boldsymbol{s}),$$

so it suffices to prove that $\sigma(\boldsymbol{y}) \in \mathcal{W}$.

Case 1. Assume $\sigma(\pi) > 0$.

By Lemma 3.2,

$$\sigma(\pi) \geqslant 1 + \frac{1}{\lambda},$$

and hence $0 < \sigma(\pi)^{-1} < 1$. Thus $\sigma(\boldsymbol{y})$ is a convex combination of $\sigma(\boldsymbol{s})$ and $\sigma(\boldsymbol{x})$. Since $\sigma(\boldsymbol{s}), \sigma(\boldsymbol{x}) \in \mathcal{W}$, the convexity of $\mathcal{W}$ implies that $\sigma(\boldsymbol{y}) \in \mathcal{W}$.

Case 2. Assume $\sigma(\pi) < 0$.

By Lemma 3.2, $\sigma(\pi) \leqslant -1 - \frac{1}{\lambda}$. Set

$$\boldsymbol{v} = \sigma(\boldsymbol{x}) - \sigma(\boldsymbol{s}), \qquad c = \sigma(\pi)^{-1} \in \left[-\frac{\lambda}{1+\lambda}, 0\right).$$

Then

$$\sigma(\boldsymbol{y}) - \sigma(\boldsymbol{s}) = c\boldsymbol{v}.$$

Since $\sigma(\boldsymbol{x}), \sigma(\boldsymbol{s}) \in \mathcal{W}$, we have

$$\mathbf{0}, \boldsymbol{v} \in \mathcal{W} - \sigma(\boldsymbol{s}).$$

By the geometric condition (7),

$$-(\mathcal{W} - \sigma(\boldsymbol{s})) \subset \left(1 + \frac{1}{\lambda}\right)(\mathcal{W} - \sigma(\boldsymbol{s})).$$

Hence

$$-\frac{\boldsymbol{v}}{1+\lambda^{-1}} \in \mathcal{W} - \sigma(\boldsymbol{s}).$$

Let

$$t = -(1+\lambda^{-1})c \in (0, 1].$$

Since $\mathcal{W} - \sigma(\boldsymbol{s})$ is convex and contains $\mathbf{0}$, we get

$$c\boldsymbol{v} = t\left(-\frac{\boldsymbol{v}}{1+\lambda^{-1}}\right) + (1-t)\mathbf{0} \in \mathcal{W} - \sigma(\boldsymbol{s}).$$

This implies that $\sigma(\boldsymbol{y}) = c\boldsymbol{v} + \sigma(\boldsymbol{s}) \in \mathcal{W}$, a contradiction to the visibility of $\boldsymbol{x}$ from $\boldsymbol{s}$. This proves

$$\Lambda_{\mathrm{vis},\{\boldsymbol{s}\}} \subseteq \left\{ \boldsymbol{x} \in \Lambda : \gcd(\boldsymbol{x} - \boldsymbol{s}) = 1,\ \sigma\left(\frac{\boldsymbol{x} - \boldsymbol{s}}{\lambda}\right) \notin \mathcal{W} - \sigma(\boldsymbol{s}) \right\}. \tag{14}$$

We now prove the reverse inclusion $\supseteq$ in (11). Suppose, toward a contradiction, that

$$\boldsymbol{x} \in \Lambda \backslash \Lambda_{\mathrm{vis},\{\boldsymbol{s}\}}, \qquad \gcd(\boldsymbol{x} - \boldsymbol{s}) = 1, \qquad \sigma\left(\frac{\boldsymbol{x} - \boldsymbol{s}}{\lambda}\right) \notin \mathcal{W} - \sigma(\boldsymbol{s}).$$

Then there exists a point $\boldsymbol{y} \in \Lambda$ that is also in the interior of the straight line segment between $\boldsymbol{s}$ and $\boldsymbol{x}$. Write $\boldsymbol{y} = \boldsymbol{s} + t(\boldsymbol{x} - \boldsymbol{s})$ with $t \in (0,1) \cap K$. Since Λ is locally finite, we may choose t minimal such that $\boldsymbol{y} \in \Lambda_{\mathrm{vis},\{\boldsymbol{s}\}}$.

Since $t \in K$, we may write $t = b/a$ with $a, b \in \mathcal{O}_K$ coprime. If b is not a unit, then

$$\boldsymbol{y} - \boldsymbol{s} = \frac{b}{a}(\boldsymbol{x} - \boldsymbol{s})$$

implies that $b \mid (\boldsymbol{y} - \boldsymbol{s})$, contradicting $\gcd(\boldsymbol{y} - \boldsymbol{s}) = 1$, which follows from the already proved inclusion (14). Thus b is a unit, and we may write $a = t^{-1} \in \mathcal{O}_K$.

If a is not a unit, then $\gcd(\boldsymbol{x} - \boldsymbol{s}) \neq 1$, which is impossible. Suppose that a is a unit. Since $t = a^{-1} \in (0,1)$, it follows that $a = \lambda^k$ for some $k \in \mathbb{N}$. Since $\boldsymbol{y} = \boldsymbol{s} + \lambda^{-k}(\boldsymbol{x} - \boldsymbol{s}) \in \Lambda$, we have

$$\sigma(\boldsymbol{y}) = \sigma(\boldsymbol{s}) + \sigma(\lambda^{-k})\sigma(\boldsymbol{x} - \boldsymbol{s}) \in \mathcal{W},$$

and hence

$$\sigma(\lambda^{-k})\sigma(\boldsymbol{x} - \boldsymbol{s}) \in \mathcal{W} - \sigma(\boldsymbol{s}). \tag{15}$$

As discussed above, for all the fields satisfying the Hammarhjelm condition, $\sigma(\lambda) = -1/\lambda$. We have

$$|\sigma(\lambda^{-k})| = (-1)^k \sigma(\lambda^{-k}) = \lambda^k \geqslant -\sigma(\lambda^{-1}) = \lambda > 1.$$

Recall that $\mathcal{W} - \sigma(\boldsymbol{s})$ is convex and contains $\mathbf{0}$. We distinguish two cases according to the parity of k.

Case 1. k is odd.

If $k = 1$ then (15) implies

$$\sigma(\lambda^{-1})\sigma(\boldsymbol{x} - \boldsymbol{s}) \in \mathcal{W} - \sigma(\boldsymbol{s}).$$

If $k \geqslant 3$, letting $\ell = \sigma(\lambda^{k-1}) \in (0,1]$, (15) implies that

$$\sigma(\lambda^{-1})\sigma(\boldsymbol{x} - \boldsymbol{s}) = \ell\sigma(\lambda^{-k})\sigma(\boldsymbol{x} - \boldsymbol{s}) \in \mathcal{W} - \sigma(\boldsymbol{s}).$$

Case 2. k is even.

Combining this with (15), we obtain

$$\sigma(\lambda^{-1})^2\sigma(\boldsymbol{x} - \boldsymbol{s}) = -\lambda\sigma(\lambda^{-1})\sigma(\boldsymbol{x} - \boldsymbol{s}) \in \mathcal{W} - \sigma(\boldsymbol{s}).$$

Equivalently,

$$\lambda\sigma(\lambda^{-1})\sigma(\boldsymbol{x} - \boldsymbol{s}) \in -(\mathcal{W} - \sigma(\boldsymbol{s})).$$

By the hypothesis

$$-(\mathcal{W} - \sigma(\boldsymbol{s})) \subset \left(1 + \frac{1}{\lambda}\right)(\mathcal{W} - \sigma(\boldsymbol{s})),$$

we get

$$\lambda\sigma(\lambda^{-1})\sigma(\boldsymbol{x} - \boldsymbol{s}) \in \left(1 + \frac{1}{\lambda}\right)(\mathcal{W} - \sigma(\boldsymbol{s})).$$

By Lemma 3.1, $1 + \frac{1}{\lambda} \leqslant \lambda$. Since $\mathcal{W} - \sigma(\boldsymbol{s})$ is convex and contains $\mathbf{0}$, the dilates are nested under positive scaling. Hence

$$\lambda\sigma(\lambda^{-1})\sigma(\boldsymbol{x} - \boldsymbol{s}) \in \lambda(\mathcal{W} - \sigma(\boldsymbol{s})),$$

and therefore

$$\sigma(\lambda^{-1})\sigma(\boldsymbol{x}-\boldsymbol{s}) \in \mathcal{W} - \sigma(\boldsymbol{s}).$$

In both cases,

$$\sigma\left(\frac{\boldsymbol{x}-\boldsymbol{s}}{\lambda}\right) \in \mathcal{W} - \sigma(\boldsymbol{s}),$$

contradicting our assumption. This proves the reverse inclusion and completes the proof. □

4. Proof of Theorem 1.1

In order to prove Theorem 1.1 we will establish a series of auxiliary lemmas. In Section 4.2 we complete the proof.

4.1. Window exclusions.

Lemma 4.1. *Let $\mathcal{W} \subseteq \mathbb{R}^d$ be a bounded, convex, and centrally symmetric set with nonempty interior, let $\boldsymbol{s} \in \Lambda(\mathcal{W}, \mathcal{L})$ satisfy the geometric condition (7). Define*

$$\mathcal{W}_{\boldsymbol{s}} \stackrel{\text{def}}{=} \frac{1}{\lambda}\big(\mathcal{W} + (\lambda+1)\sigma(\boldsymbol{s})\big).$$

Then

$$\mathcal{W}_{\boldsymbol{s}} \subseteq \mathcal{W}.$$

Proof. Since $\boldsymbol{s} \in \Lambda(\mathcal{W}, \mathcal{L})$, we have $\sigma(\boldsymbol{s}) \in \mathcal{W}$. Hence

$$\mathbf{0} \in \mathcal{W} - \sigma(\boldsymbol{s}).$$

Since $\mathcal{W}$ is convex, the translated set $\mathcal{W} - \sigma(\boldsymbol{s})$ is also convex. Therefore, because it contains $\mathbf{0}$, we have

$$t(\mathcal{W} - \sigma(\boldsymbol{s})) \subseteq \mathcal{W} - \sigma(\boldsymbol{s}) \text{ for every } t \in [0,1].$$

Let $\boldsymbol{y} \in \mathcal{W}_{\boldsymbol{s}}$. By definition, there exists $\boldsymbol{w} \in \mathcal{W}$ such that

$$\boldsymbol{y} = \frac{\boldsymbol{w} + (\lambda+1)\sigma(\boldsymbol{s})}{\lambda} = \sigma(\boldsymbol{s}) + \frac{\boldsymbol{w} + \sigma(\boldsymbol{s})}{\lambda}.$$

Thus

$$\boldsymbol{y} - \sigma(\boldsymbol{s}) = \frac{\boldsymbol{w} + \sigma(\boldsymbol{s})}{\lambda}.$$

Since $\mathcal{W}$ is centrally symmetric, $-\boldsymbol{w} \in \mathcal{W}$. Hence

$$\boldsymbol{y} - \sigma(\boldsymbol{s}) = -\frac{1}{\lambda}\big((-\boldsymbol{w}) - \sigma(\boldsymbol{s})\big) \in -\frac{1}{\lambda}(\mathcal{W} - \sigma(\boldsymbol{s})).$$

By hypothesis,

$$-(\mathcal{W} - \sigma(\boldsymbol{s})) \subseteq \left(1 + \frac{1}{\lambda}\right)(\mathcal{W} - \sigma(\boldsymbol{s})).$$

Multiplying by $1/\lambda$, we obtain

$$-\frac{1}{\lambda}(\mathcal{W} - \sigma(\boldsymbol{s})) \subseteq \frac{1+\lambda^{-1}}{\lambda}(\mathcal{W} - \sigma(\boldsymbol{s})).$$

Using Lemma 3.1, we get

$$0 < \frac{1+\lambda^{-1}}{\lambda} \leqslant 1.$$

Since $\mathcal{W} - \sigma(\boldsymbol{s})$ is convex and contains $\mathbf{0}$, it follows that

$$\frac{1+\lambda^{-1}}{\lambda}(\mathcal{W} - \sigma(\boldsymbol{s})) \subseteq \mathcal{W} - \sigma(\boldsymbol{s}).$$

Therefore

$$\boldsymbol{y} - \sigma(\boldsymbol{s}) \in \mathcal{W} - \sigma(\boldsymbol{s}),$$

and hence $\boldsymbol{y} \in \mathcal{W}$. Since $\boldsymbol{y} \in \mathcal{W}_{\boldsymbol{s}}$ was arbitrary, we conclude that

$$\mathcal{W}_{\boldsymbol{s}} \subseteq \mathcal{W}.$$

□

We denote the set of $\boldsymbol{s}$-primitive points in $\Lambda(\mathcal{W}_{\boldsymbol{s}}, \mathcal{L})$ by

$$\Lambda_{\mathrm{pr},\{\boldsymbol{s}\}}(\mathcal{W}_{\boldsymbol{s}}, \mathcal{L}) \stackrel{\mathrm{def}}{=} \{\boldsymbol{x} \in \Lambda(\mathcal{W}_{\boldsymbol{s}}, \mathcal{L}) : \gcd(\boldsymbol{x} - \boldsymbol{s}) = 1\}.$$

More generally, if $\mathcal{W}' \subset \mathcal{W}$, we denote the set of S-primitive points in $\Lambda(\mathcal{W}', \mathcal{L})$ by

$$\Lambda_{\mathrm{pr},S}(\mathcal{W}', \mathcal{L}) \stackrel{\mathrm{def}}{=} \left\{\boldsymbol{x} \in \Lambda(\mathcal{W}', \mathcal{L}) : \gcd(\boldsymbol{x} - \boldsymbol{s}) = 1 \text{ for all } \boldsymbol{s} \in S\right\}.$$

By Lemmas 2.2 and 4.1, we have

$$\Lambda_{\mathrm{vis},S} = \Lambda_{\mathrm{pr},S}(\mathcal{W}, \mathcal{L}) \setminus \bigcup_{\boldsymbol{s} \in S} \left(\Lambda_{\mathrm{pr},\{\boldsymbol{s}\}}(\mathcal{W}_{\boldsymbol{s}}, \mathcal{L}) \cap \Lambda_{\mathrm{pr},S}(\mathcal{W}, \mathcal{L})\right). \tag{16}$$

Applying the inclusion-exclusion principle to the union in (16), we obtain

$$\begin{aligned} \#(\Lambda_{\mathrm{vis},S} \cap TD) &= \#(\Lambda_{\mathrm{pr},S}(\mathcal{W}, \mathcal{L}) \cap TD) \\ &\quad - \sum_{\varnothing \neq I \subseteq S} (-1)^{\#(I)+1} \#\left(\Lambda_{\mathrm{pr},S}(\mathcal{W}, \mathcal{L}) \cap \bigcap_{\boldsymbol{s} \in I} \Lambda_{\mathrm{pr},\{\boldsymbol{s}\}}(\mathcal{W}_{\boldsymbol{s}}, \mathcal{L}) \cap TD\right). \end{aligned} \tag{17}$$

Lemma 4.2. *Let $\mathcal{W} \subseteq \mathbb{R}^d$ be a bounded, convex, and centrally symmetric set with nonempty interior, let $\boldsymbol{s} \in \Lambda(\mathcal{W}, \mathcal{L})$ satisfying the condition (7). Let $I \subseteq S$ be nonempty. Then*

$$\Lambda_{\mathrm{pr},S}(\mathcal{W}, \mathcal{L}) \cap \bigcap_{\boldsymbol{s} \in I} \Lambda_{\mathrm{pr},\{\boldsymbol{s}\}}(\mathcal{W}_{\boldsymbol{s}}, \mathcal{L}) = \Lambda_{\mathrm{pr},S}(\mathcal{W}_I, \mathcal{L}).$$

Proof. Let $\boldsymbol{x} \in \Lambda_{\mathrm{pr},S}(\mathcal{W}, \mathcal{L}) \cap \bigcap_{\boldsymbol{s} \in I} \Lambda_{\mathrm{pr},\{\boldsymbol{s}\}}(\mathcal{W}_{\boldsymbol{s}}, \mathcal{L})$. Since $\boldsymbol{x} \in \Lambda_{\mathrm{pr},S}(\mathcal{W}, \mathcal{L})$, we have

$$\gcd(\boldsymbol{x} - \boldsymbol{s}) = 1 \qquad \text{for all } \boldsymbol{s} \in S.$$

Moreover, for every $\boldsymbol{s} \in I$, the condition $\boldsymbol{x} \in \Lambda_{\mathrm{pr},\{\boldsymbol{s}\}}(\mathcal{W}_{\boldsymbol{s}}, \mathcal{L})$ implies $\boldsymbol{x} \in \Lambda(\mathcal{W}_{\boldsymbol{s}}, \mathcal{L})$. Hence

$$\sigma(\boldsymbol{x}) \in \bigcap_{\boldsymbol{s} \in I} \mathcal{W}_{\boldsymbol{s}} = \mathcal{W}_I,$$

and therefore $\boldsymbol{x} \in \Lambda(\mathcal{W}_I, \mathcal{L})$. Combining this with the coprimality conditions for all $\boldsymbol{s} \in S$, we obtain

$$\boldsymbol{x} \in \Lambda_{\mathrm{pr},S}(\mathcal{W}_I, \mathcal{L}).$$

Thus

$$\Lambda_{\mathrm{pr},S}(\mathcal{W}, \mathcal{L}) \cap \bigcap_{\boldsymbol{s} \in I} \Lambda_{\mathrm{pr},\{\boldsymbol{s}\}}(\mathcal{W}_{\boldsymbol{s}}, \mathcal{L}) \subseteq \Lambda_{\mathrm{pr},S}(\mathcal{W}_I, \mathcal{L}).$$

Conversely, let $\boldsymbol{x} \in \Lambda_{\mathrm{pr},S}(\mathcal{W}_I, \mathcal{L})$. Then

$$\sigma(\boldsymbol{x}) \in \mathcal{W}_I = \bigcap_{\boldsymbol{s} \in I} \mathcal{W}_{\boldsymbol{s}}$$

and

$$\gcd(\boldsymbol{x} - \boldsymbol{s}) = 1 \qquad \text{for all } \boldsymbol{s} \in S.$$

Since $\mathcal{W}_I \subseteq \mathcal{W}_{\boldsymbol{s}}$ for every $\boldsymbol{s} \in I$, it follows that

$$\boldsymbol{x} \in \Lambda_{\mathrm{pr},\{\boldsymbol{s}\}}(\mathcal{W}_{\boldsymbol{s}}, \mathcal{L}) \qquad \text{for every } \boldsymbol{s} \in I.$$

Also, by Lemma 4.1, we have $\mathcal{W}_{\boldsymbol{s}} \subseteq \mathcal{W}$ for every $\boldsymbol{s} \in I$, and hence

$$\mathcal{W}_I \subseteq \mathcal{W}.$$

Therefore $\boldsymbol{x} \in \Lambda(\mathcal{W}, \mathcal{L})$. Together with the coprimality conditions for all $\boldsymbol{s} \in S$, this gives

$$\boldsymbol{x} \in \Lambda_{\mathrm{pr},S}(\mathcal{W}, \mathcal{L}).$$

Thus

$$\Lambda_{\mathrm{pr},S}(\mathcal{W}_I,\mathcal{L}) \subseteq \Lambda_{\mathrm{pr},S}(\mathcal{W},\mathcal{L}) \cap \bigcap_{\boldsymbol{s}\in I} \Lambda_{\mathrm{pr},\{\boldsymbol{s}\}}(\mathcal{W}_{\boldsymbol{s}},\mathcal{L}),$$

and the proof is complete. □

Using (17) and Lemma 4.2, we obtain the following corollary.

Corollary 4.3. *Under the hypotheses of Theorem 1.1, for every sufficiently large $T>0$,*

$$\begin{aligned}\#(\Lambda_{\mathrm{vis},S}(\mathcal{W},\mathcal{L})\cap TD) = &\#(\Lambda_{\mathrm{pr},S}(\mathcal{W},\mathcal{L})\cap TD)\\ &- \sum_{\varnothing\neq I\subseteq S} (-1)^{\#(I)+1}\,\#(\Lambda_{\mathrm{pr},S}(\mathcal{W}_I,\mathcal{L})\cap TD)\,.\end{aligned}$$

The final step in the proof of Theorem 1.1 is the following lemma.

Lemma 4.4. *Under the hypotheses of Theorem 1.1, let either $\mathcal{W}'=\mathcal{W}$, or $\mathcal{W}'=\mathcal{W}_I$ for some nonempty subset $I\subseteq S$. Then the density of $\Lambda_{\mathrm{pr},S}(\mathcal{W}',\mathcal{L})$ along $\{TD\}_{T>0}$ exists and is given by:*

$$\theta\big(\Lambda_{\mathrm{pr},S}(\mathcal{W}',\mathcal{L})\big) = \frac{\mathrm{vol}(\mathcal{W}')}{\mathrm{covol}(\mathcal{L})}\prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}}\left(1-\frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right).$$

4.2. **Completion of the proof of Theorem 1.1.** We assume Lemma 4.4 for now and postpone its proof to the next subsection. Combining Corollary 4.3 with Lemma 4.4, we obtain that the density of $\Lambda_{\mathrm{vis},S}(\mathcal{W},\mathcal{L})$ along the sequence $\{TD\}_{T>0}$ exists. Moreover,

$$\begin{aligned}\theta(\Lambda_{\mathrm{vis},S}(\mathcal{W},\mathcal{L})) &= \lim_{T\to\infty}\frac{\#(\Lambda_{\mathrm{pr},S}(\mathcal{W},\mathcal{L})\cap TD)}{\mathrm{vol}(TD)}\\ &\quad - \sum_{\varnothing\neq I\subseteq S}(-1)^{\#(I)+1}\lim_{T\to\infty}\frac{\#(\Lambda_{\mathrm{pr},S}(\mathcal{W}_I,\mathcal{L})\cap TD)}{\mathrm{vol}(TD))}\\ &= \frac{\mathrm{vol}(\mathcal{W})}{\mathrm{covol}(\mathcal{L})}\prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}}\left(1-\frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right)\\ &\quad - \sum_{\varnothing\neq I\subseteq S}(-1)^{\#(I)+1}\frac{\mathrm{vol}(\mathcal{W}_I)}{\mathrm{covol}(\mathcal{L})}\prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}}\left(1-\frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right)\\ &= \mathrm{dens}(\mathcal{L})\left(\mathrm{vol}(\mathcal{W})-\sum_{\varnothing\neq I\subseteq S}(-1)^{\#(I)+1}\mathrm{vol}(\mathcal{W}_I)\right)\prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}}\left(1-\frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right).\end{aligned}$$

This is the formula claimed in Theorem 1.1.

It remains to prove Lemma 4.4.

4.3. **Proof of Lemma 4.4.** For a rational prime p, let M_p denote the set of prime ideals of $\mathcal{O}_K$ lying above p. Since K is quadratic, we have

$$\#(M_p)\leqslant 2.$$

Let p_i denote the ith rational prime, and define

$$\mathcal{E}_t \overset{\mathrm{def}}{=} \bigcup_{i=1}^{t} M_{p_i}, \qquad t\in\mathbb{N}.$$

For a prime ideal $\mathfrak{p}$, define

$$\mathfrak{p}^d \overset{\mathrm{def}}{=} \{(x_1,\dots,x_d)\in\mathcal{O}_K^d : x_i\in\mathfrak{p}\}.$$

For $\mathcal{W}'$ as in Lemma 4.4, define

$$\Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}',\mathcal{L}) \overset{\mathrm{def}}{=} \Big\{\boldsymbol{x}\in\Lambda(\mathcal{W}',\mathcal{L}) : \boldsymbol{x}-\boldsymbol{s}\notin\mathfrak{p}^d \text{ for all } \mathfrak{p}\in\mathcal{E}_t \text{ and all } \boldsymbol{s}\in S\Big\}.$$

Equivalently, $\Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}',\mathcal{L})$ is obtained from $\Lambda(\mathcal{W}',\mathcal{L})$ by imposing only the primitivity conditions coming from prime ideals above the first t rational primes.

In the following lemma, we show that the density of $\Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}',\mathcal{L})$ along averaging sets TD exists.

Lemma 4.5. *Under the hypotheses of Lemma 4.4, the density of* $\Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}',\mathcal{L})$ *along the sequence* $\{TD\}_{T>0}$ *exists. Moreover,*

$$\theta(\Lambda_{\mathrm{pr},S,\mathcal{E}_t}) = \frac{\mathrm{vol}(\mathcal{W}')}{\mathrm{covol}(\mathcal{L})} \cdot \prod_{\mathfrak{p}\in\mathcal{E}_t} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right). \tag{18}$$

Proof. Let $G = \mathbb{R}^d$ be the physical space, and let the internal space be given by

$$H = \mathbb{R}^d \times \prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}} H_{\mathfrak{p}},$$

where $H_{\mathfrak{p}} = \left(\mathcal{O}_K/\mathfrak{p}\right)^d$. In the group H, each finite factor is equipped with the normalized counting measure. Both G and H are σ-compact and LCAGs, and the group H is compactly generated. Let $\mathcal{L}''$ be the image of the diagonal embedding

$$\mathcal{O}_K^d \hookrightarrow G \times H, \qquad \boldsymbol{x} \hookrightarrow (\boldsymbol{x}, \sigma(\boldsymbol{x}), (\pi_{\mathfrak{p}}(\boldsymbol{x}))_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}}).$$

The restriction $\pi''_{\mathrm{phys}}|_{\mathcal{L}''} : \mathcal{L}'' \to G$ is injective. Using the Chinese remainder theorem and the fact that $\pi_{\mathrm{int}}(\mathcal{L})$ is dense in $\mathbb{R}^d$ it can be shown that the image $\pi''_{\mathrm{int}}(\mathcal{L}'')$ is dense in H. The projection

$$G \times H \longrightarrow \mathbb{R}^{2d}, \qquad (\boldsymbol{u}, \boldsymbol{v}, (\boldsymbol{w}_{\mathfrak{p}})_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}}) \longmapsto (\boldsymbol{u}, \boldsymbol{v})$$

descends to a well-defined continuous surjective map

$$(G \times H)/\mathcal{L}'' \longrightarrow \mathbb{R}^{2d}\Big/\mathcal{L}$$

whose kernel is $\prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}} H_{\mathfrak{p}}$. Since $\mathcal{L}$ is a cocompact lattice, $G \times H/\mathcal{L}''$ is compact, which shows that $\mathcal{L}''$ is a cocompact lattice.

Define

$$\mathcal{W}'' \stackrel{\mathrm{def}}{=} \mathcal{W}' \times \prod_{\mathfrak{p}\in\mathcal{E}_t} \mathcal{W}_{\mathfrak{p}} \times \prod_{\substack{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K} \\ \mathfrak{p}\notin\mathcal{E}_t}} H_{\mathfrak{p}},$$

where

$$\mathcal{W}_{\mathfrak{p}} \stackrel{\mathrm{def}}{=} \left(\mathcal{O}_K/\mathfrak{p}\right)^d \setminus \pi_{\mathfrak{p}}(S).$$

By construction, the associated cut-and-project set $\Lambda(\mathcal{W}'',\mathcal{L}'')$ consists exactly of those $\boldsymbol{x} \in \mathcal{O}_K^d$ such that

$$\sigma(\boldsymbol{x}) \in \mathcal{W}' \qquad \text{and} \qquad \pi_{\mathfrak{p}}(\boldsymbol{x}) \notin \pi_{\mathfrak{p}}(S) \quad \text{for every } \mathfrak{p} \in \mathcal{E}_t.$$

Equivalently,

$$\Lambda(\mathcal{W}'',\mathcal{L}'') = \Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}',\mathcal{L}).$$

Suppose that for a prime ideal $\mathfrak{p} \in \mathcal{E}_t$, we have $s(\mathfrak{p}) = N(\mathfrak{p})^d$. In this case $\mathcal{W}'' = \varnothing$. Therefore, $\Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}',\mathcal{L}) = \varnothing$. Since $\Lambda_{\mathrm{pr},S}(\mathcal{W}',\mathcal{L}) \subseteq \Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}',\mathcal{L})$, it follows that (18) holds true.

Now suppose that $s(\mathfrak{p}) < N(\mathfrak{p})^d$ for all $\mathfrak{p} \in \mathcal{E}_t$. In this case, the set $\mathcal{W}''$ is a regular window: it is relatively compact, has nonempty interior, and its boundary has Haar measure zero. Furthermore,

$$\mathrm{vol}(\mathcal{W}'') = \mathrm{vol}(\mathcal{W}') \prod_{\mathfrak{p}\in\mathcal{E}_t} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right).$$

It follows from the construction that $\Lambda(\mathcal{W}'', \mathcal{L}'')$ is an irreducible cut-and-project set. Applying the density formula (9) for irreducible cut-and-project sets, we get

$$\begin{aligned}\theta(\Lambda_{\mathrm{pr},S,\mathcal{E}_t}) = \theta(\Lambda(\mathcal{W}'', \mathcal{L}'')) &= \frac{\mathrm{vol}(\mathcal{W}')}{\mathrm{covol}(\mathcal{L}'')} \prod_{\mathfrak{p}\in\mathcal{E}_t} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right) \\ &= \frac{\mathrm{vol}(\mathcal{W}')}{\mathrm{covol}(\mathcal{L})} \prod_{\mathfrak{p}\in\mathcal{E}_t} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right)\end{aligned}$$

completing the proof of Lemma 4.5. □

Since

$$\Lambda_{\mathrm{pr},S}(\mathcal{W}', \mathcal{L}) \subseteq \Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}', \mathcal{L})$$

for every $t \in \mathbb{N}$, Lemma 4.5 gives

$$\overline{\theta}\big(\Lambda_{\mathrm{pr},S}(\mathcal{W}', \mathcal{L})\big) \leqslant \theta\big(\Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}', \mathcal{L})\big) = \frac{\mathrm{vol}(\mathcal{W}')}{\mathrm{covol}(\mathcal{L})} \prod_{\mathfrak{p}\in\mathcal{E}_t} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right).$$

Letting $t \to \infty$, we obtain

$$\overline{\theta}\big(\Lambda_{\mathrm{pr},S}(\mathcal{W}', \mathcal{L})\big) \leqslant \frac{\mathrm{vol}(\mathcal{W}')}{\mathrm{covol}(\mathcal{L})} \prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right).$$

We will make use of the following statement in the sequel.

Lemma 4.6. *Suppose that $B \subseteq A \subset \mathbb{R}^d$ are nonempty sets and $\theta(A)$ exists along an averaging sequence $\{D_T\}_{T>0}$. Then,*

$$\underline{\theta}\big(B\big) = \theta\big(A\big) - \overline{\theta}\left(A\backslash B\right).$$

Observe that for $B = \Lambda_{\mathrm{pr},S}(\mathcal{W}', \mathcal{L})$ and $A = \Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}', \mathcal{L})$, we have $B \subseteq A$ and $\theta(A)$ exists by Lemma 4.5 along the averaging sequence $\{TD\}_{T>0}$. Then Lemma 4.6 implies the following.

Corollary 4.7. *For every $t \in \mathbb{N}$,*

$$\underline{\theta}\big(\Lambda_{\mathrm{pr},S}(\mathcal{W}', \mathcal{L})\big) = \theta\big(\Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}', \mathcal{L})\big) - \overline{\theta}\left(\Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}', \mathcal{L})\backslash\Lambda_{\mathrm{pr},S}(\mathcal{W}', \mathcal{L})\right).$$

Proof of Lemma 4.6. Because $B \subseteq A$, we can decompose A into the disjoint union

$$A = B \sqcup (A\backslash B).$$

For $T > 0$, we have

$$\frac{\#(B \cap D_T)}{\mathrm{vol}(D_T)} = \frac{\#(A \cap D_T)}{\mathrm{vol}(D_T)} - \frac{\#((A\backslash B) \cap D_T)}{\mathrm{vol}(D_T)}.$$

Let

$$a_T \overset{\text{def}}{=} \frac{\#(A \cap D_T)}{\mathrm{vol}(D_T)} \quad \text{and} \quad c_T \overset{\text{def}}{=} \frac{\#((A\backslash B) \cap D_T)}{\mathrm{vol}(D_T)}.$$

Let T_n be a sequence such that $T_n \xrightarrow{n\to\infty} \infty$ and

$$\lim_{n\to\infty} c_{T_n} = \limsup_{T\to\infty} c_T.$$

Hence $\lim_{n\to\infty} a_{T_n} = \theta(A)$. On one hand

$$\theta(A) - \limsup_{T\to\infty} c_T = \lim_{n\to\infty} a_{T_n} - \lim_{n\to\infty} c_{T_n} = \lim_{n\to\infty} (a_{T_n} - c_{T_n}) \geqslant \liminf_{T\to\infty} (a_T - c_T).$$

On the other hand

$$\liminf_{T\to\infty} (a_T - c_T) \geqslant \liminf_{T\to\infty} a_T + \liminf_{T\to\infty} (-c_T) = \theta(A) - \limsup_{T\to\infty} c_T.$$

Combining the previous inequalities we obtain $\underline{\theta}(B) = \theta(A) - \overline{\theta}(A\backslash B)$. □

We have the following intermediate result.

Lemma 4.8. *Under the hypotheses of Lemma 4.4,*

$$\lim_{t\to\infty} \overline{\theta}\left(\Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}',\mathcal{L})\backslash\Lambda_{\mathrm{pr},S}(\mathcal{W}',\mathcal{L})\right) = 0.$$

The proof of Lemma 4.4 follows from Lemma 4.8 in combination with Lemma 4.5 and Corollary 4.7.

Proof of Lemma 4.8. Define

$$A_t \stackrel{\text{def}}{=} \Lambda_{\mathrm{pr},S,\mathcal{E}_t}(\mathcal{W}',\mathcal{L})\backslash\Lambda_{\mathrm{pr},S}(\mathcal{W}',\mathcal{L}).$$

Thus, A_t consists of those points that satisfy the primitivity conditions for every $\boldsymbol{s}\in S$ and every prime ideal $\mathfrak{p}\in\mathcal{E}_t$. However, they fail the primitivity condition for at least one pair $(s,\mathfrak{p})$, where $\boldsymbol{s}\in S$ and $\mathfrak{p}\notin\mathcal{E}_t$.

For $g\in\mathcal{O}_K$, let a_g be the diagonal matrix

$$a_g \stackrel{\text{def}}{=} \mathrm{diag}(\underbrace{g,\dots,g}_{d\text{ times}},\underbrace{\sigma(g),\dots,\sigma(g)}_{d\text{ times}}),$$

and define

$$\mathcal{L}_{(g)} \stackrel{\text{def}}{=} a_g\mathcal{L} = \{(g\boldsymbol{x},\sigma(g\boldsymbol{x})) : \boldsymbol{x}\in\mathcal{O}_K^d\}.$$

Since $\mathcal{O}_K$ is a ring, $\mathcal{L}_{(g)}\subseteq\mathcal{L}$, and

$$\mathrm{covol}(\mathcal{L}_{(g)}) = |N(g)|^d\mathrm{covol}(\mathcal{L}).$$

As $\mathcal{O}_K$ is a principal ideal domain, if $\mathfrak{p}=(g)$ we write $\mathcal{L}_\mathfrak{p}$ for $\mathcal{L}_{(g)}$. We claim that

$$A_t \subseteq \bigcup_{\boldsymbol{s}\in S}\bigcup_{\substack{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}\\ \mathfrak{p}\notin\mathcal{E}_t}}\left(\boldsymbol{s}+\Lambda(\mathcal{W}'-\sigma(\boldsymbol{s}),\mathcal{L}_\mathfrak{p})\right). \tag{19}$$

Indeed, let $\boldsymbol{x}\in A_t$ be an arbitrary point. Then there exist $\boldsymbol{s}_0\in S$ and a prime ideal $\mathfrak{p}\notin\mathcal{E}_t$ such that

$$\mathfrak{p}\;\Big|\;(\boldsymbol{x}-\boldsymbol{s}_0),$$

or equivalently $\boldsymbol{x}-\boldsymbol{s}_0\in\mathfrak{p}^d$. Since $\sigma(\boldsymbol{x})\in\mathcal{W}'$, we have

$$\sigma(\boldsymbol{x}-\boldsymbol{s}_0)\in\mathcal{W}'-\sigma(\boldsymbol{s}_0).$$

Thus

$$\boldsymbol{x}\in\boldsymbol{s}_0+\Lambda(\mathcal{W}'-\sigma(\boldsymbol{s}_0),\mathcal{L}_\mathfrak{p}),$$

which proves (19).

Claim 1. *There exists a constant $c>0$, such that for every $t\in\mathbb{N}$ and every $T>p_t^2$,*

$$A_t\cap TD \subseteq \bigcup_{\boldsymbol{s}\in S}\bigcup_{\substack{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}\\ \mathfrak{p}\notin\mathcal{E}_t\\ N(\mathfrak{p})\leqslant cT}}\left(\boldsymbol{s}+\Lambda(\mathcal{W}'-\sigma(\boldsymbol{s}),\mathcal{L}_\mathfrak{p})\cap TD\right). \tag{20}$$

That is, the union on the right-hand side of (20) is taken over finitely many prime ideals $\mathfrak{p}\notin\mathcal{E}_t$ with norm at most cT.

Proof of claim 1. Fix $\boldsymbol{s}\in S$, and suppose

$$\boldsymbol{x}\in(\boldsymbol{s}+\Lambda(\mathcal{W}'-\sigma(\boldsymbol{s}),\mathcal{L}_\mathfrak{p}))\cap TD,\qquad \boldsymbol{x}\neq\boldsymbol{s}.$$

Then $\boldsymbol{x}-\boldsymbol{s}\in\mathfrak{p}^d$. Choose a coordinate j such that $x_j-s_j\neq 0$. Because $x_j-s_j\in\mathfrak{p}$, we have $(x_j-s_j)\subseteq\mathfrak{p}$, where (x_i-s_j) is the ideal generated by x_j-s_j. For ideals, inclusion reverses the norm inequality; hence

$$N(\mathfrak{p})\leqslant N((x_j-s_j)) = |N(x_j-s_j)|.$$

Now $N(x_j - s_j) = (x_j - s_j)\sigma(x_j - s_j)$. Since D is bounded and $\boldsymbol{x} \in TD$, there is a constant $C_D > 0$ depending only on D, such that $|x_j| \leqslant C_D T$. Since S is finite, we obtain

$$|x_j - s_j| \leqslant c_1 T, \qquad \forall\, 1 \leqslant j \leqslant d,\ \forall\, \boldsymbol{s} \in S, \tag{21}$$

where c_1 is a constant independent of $\mathfrak{p}, \boldsymbol{s}, T$. Also, $\sigma(x_j - s_j) = \sigma(x_j) - \sigma(s_j)$. The window $\mathcal{W}'$ is bounded. Thus, given $\boldsymbol{y} = (y_1, \ldots, y_d) \in \mathcal{W}' - \sigma(\boldsymbol{s})$, there exists $c_{\boldsymbol{s}} > 0$ such that

$$|y_j| \leqslant c_{\boldsymbol{s}}, \qquad 1 \leqslant j \leqslant d.$$

The constant $c_2 \overset{\text{def}}{=} \max_{\boldsymbol{s} \in S}\{c_{\boldsymbol{s}}\} > 0$ is independent of $\boldsymbol{x}, \boldsymbol{s}, T$ and $\mathfrak{p}$, and satisfies

$$|\sigma(x_j - s_j)| \leqslant c_2. \tag{22}$$

Combining (21) and (22), we obtain

$$N(\mathfrak{p}) \leqslant c_1 c_2 T.$$

Let H_n be the number of ideals in $\mathcal{O}_K$ of norm n. It follows from [Mar18, Theorem 39] that

$$H_n \leqslant H \cdot n^{1/2},$$

where $H > 0$ is a constant independent of n. Hence, only finitely many prime ideals can occur in (20). This completes the proof of the claim. ▲

For $\boldsymbol{s} \in S$ and $\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}$, write

$$N_{\boldsymbol{s},\mathfrak{p}}(T) \overset{\text{def}}{=} \#\big(\big(\boldsymbol{s} + \Lambda(\mathcal{W}' - \sigma(\boldsymbol{s}), \mathcal{L}_{\mathfrak{p}})\big) \cap TD\big)\,.$$

By (20), we have

$$\#(A_t \cap TD) \leqslant \sum_{s \in S} \sum_{\mathfrak{p} \notin \mathcal{E}_t} N_{s,\mathfrak{p}}(T), \tag{23}$$

where the second sum runs over finitely many primes. By [GKW25, Lemma 5.4] we have for $N(\mathfrak{p}) \leqslant T$:

$$\begin{aligned} N_{\boldsymbol{s},\mathfrak{p}}(T) &\leqslant \frac{\mathrm{vol}(TD) \cdot \mathrm{vol}(\mathcal{W}')}{\mathrm{covol}(\mathcal{L}_{\mathfrak{p}})} + C_1 \left(\frac{T}{N(\mathfrak{p})}\right)^{d-\frac{1}{2}} \\ N_{\boldsymbol{s},\mathfrak{p}}(T) &\leqslant C_0 \frac{T^d}{N(\mathfrak{p})^d} + C_1 \left(\frac{T}{N(\mathfrak{p})}\right)^{d-\frac{1}{2}} \\ \frac{N_{\boldsymbol{s},\mathfrak{p}}(T)}{\mathrm{vol}(TD)} &\leqslant \frac{C_2}{N(\mathfrak{p})^d} + \frac{C_3}{T^{1/2} N(\mathfrak{p})^{d-\frac{1}{2}}}, \end{aligned} \tag{24}$$

where $C_0, C_1, C_2, C_3 > 0$ are constants independent of $\boldsymbol{s}, \mathfrak{p}$ and T. It follows from [Ham22, Lemma 4.6] (see, also [GKW25, Lemma 5.5]), that for prime ideals satisfying $N(\mathfrak{p}) > T$

$$\frac{N_{\boldsymbol{s},\mathfrak{p}}(T)}{\mathrm{vol}(TD)} \leqslant \frac{C_4}{N(\mathfrak{p})^d}, \tag{25}$$

for some constant C_4 independent of $\boldsymbol{s}, \mathfrak{p}$ and T. Splitting $N_{\boldsymbol{s},\mathfrak{p}}(T)$ over sums of primes according to $N(\mathfrak{p}) \leqslant T$ or $N(\mathfrak{p}) > T$, we obtain

$$N_{\boldsymbol{s},\mathfrak{p}}(T) = \sum_{\substack{\mathfrak{p} \notin \mathcal{E}_t \\ N(\mathfrak{p}) \leqslant T}} N_{s,\mathfrak{p}}(T) \quad + \sum_{\substack{\mathfrak{p} \notin \mathcal{E}_t \\ T < N(\mathfrak{p}) \leqslant c_1 c_2 T}} N_{s,\mathfrak{p}}(T).$$

By (23), (24) and (25), there is a constant C independent of $\boldsymbol{s}$, $\mathfrak{p}$, and T, such that

$$\frac{\#(A_t \cap TD)}{\mathrm{vol}(TD)} \leqslant C \cdot \#(S) \sum_{\substack{\mathfrak{p}\notin\mathcal{E}_t \\ N(\mathfrak{p})\leqslant T}} \frac{1}{N(\mathfrak{p})^d} + \frac{C\cdot\#(S)}{T^{1/2}} \sum_{\substack{\mathfrak{p}\notin\mathcal{E}_t \\ N(\mathfrak{p})\leqslant T}} \frac{1}{N(\mathfrak{p})^{d-\frac{1}{2}}} + C\cdot\#(S) \sum_{\substack{\mathfrak{p}\notin\mathcal{E}_t \\ N(\mathfrak{p})> T}} \frac{1}{N(\mathfrak{p})^d}.$$

Combining the first and third sums yields:

$$\frac{\#(A_t \cap TD)}{\mathrm{vol}(TD)} \leqslant C \cdot \#(S) \sum_{\substack{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K} \\ \mathfrak{p}\notin\mathcal{E}_t}} \frac{1}{N(\mathfrak{p})^d} + \frac{C\cdot\#(S)}{T^{1/2}} \sum_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}} \frac{1}{N(\mathfrak{p})^{d-\frac{1}{2}}}. \tag{26}$$

Because $d \geqslant 2$, we have $d - \frac{1}{2} > 1$, and hence,

$$\sum_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}} \frac{1}{N(\mathfrak{p})^{d-\frac{1}{2}}} < \infty.$$

Therefore, for fixed t and taking $\limsup_{T\to\infty}$ the second term in (26) vanishes. Thus, the upper density satisfies:

$$\overline{\theta}(A_t) \leqslant C \cdot \#(S) \sum_{\substack{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K} \\ \mathfrak{p}\notin\mathcal{E}_t}} \frac{1}{N(\mathfrak{p})^d}. \tag{27}$$

Finally, since $d \geqslant 2$, $\sum_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}} \frac{1}{N(\mathfrak{p})^d} < \infty$. The sets $\mathcal{E}_t$ exhaust $\mathcal{P}_{\mathcal{O}_K}$ as $t \to \infty$,

$$\sum_{\substack{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K} \\ \mathfrak{p}\notin\mathcal{E}_t}} \frac{1}{N(\mathfrak{p})^d} \xrightarrow[t\to\infty]{} 0. \tag{28}$$

Combining (27) and (28), we conclude that

$$\lim_{t\to\infty} \overline{\theta}(A_t) = 0$$

proving Lemma 4.8. □

5. Proof of Corollary 1.2

Let $G = \mathbb{R}^d$ be the physical space, and let

$$H = \mathbb{R}^d \times \prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}} H_{\mathfrak{p}}$$

be the internal space as in the proof of Theorem 1.1. Let $\mathcal{L}_2$ be the image of the diagonal embedding

$$\mathcal{O}_K^d \hookrightarrow G \times H, \quad \boldsymbol{x} \hookrightarrow \left(\boldsymbol{x}, \sigma(\boldsymbol{x}), (\pi_{\mathfrak{p}}(\boldsymbol{x}))_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}}\right).$$

Then $\mathcal{L}_2$ is a lattice. The projection of $\mathcal{L}_2$ to G is injective, while the projection to H is dense. Moreover, $\mathrm{covol}(\mathcal{L}_2) = \mathrm{covol}(\mathcal{L})$. Let

$$\mathcal{W}_2 = \mathcal{W} \setminus \left(\bigcup_{\boldsymbol{s}\in S} \mathcal{W}_{\boldsymbol{s}}\right) \times \prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}} \mathcal{W}_{\mathfrak{p}},$$

where

$$\mathcal{W}_{\mathfrak{p}} \overset{\text{def}}{=} \left(\mathcal{O}_K/\mathfrak{p}\right)^d \setminus \pi_{\mathfrak{p}}(S).$$

We claim that

$$\Lambda_{\mathrm{vis},S} = \Lambda(\mathcal{W}_2, \mathcal{L}_2).$$

In fact, by the definitions of $\Lambda_{\mathrm{vis},S}$, $\Lambda_{\mathrm{vis},\boldsymbol{s}}$, and Lemma 2.2, we have

$$\begin{aligned}
\boldsymbol{x} \in \Lambda_{\mathrm{vis},S} &\iff \boldsymbol{x} \in \Lambda_{\mathrm{vis},\boldsymbol{s}}, \qquad \forall \boldsymbol{s} \in S \\
&\iff \sigma(\boldsymbol{x}) \in \mathcal{W},\ \sigma(\boldsymbol{x}) \notin \mathcal{W}_{\boldsymbol{s}},\ \gcd(\boldsymbol{x}-\boldsymbol{s}) = 1, \qquad \forall \boldsymbol{s} \in S \\
&\iff \sigma(\boldsymbol{x}) \in \mathcal{W} \backslash \bigcup_{\boldsymbol{s}\in S} \mathcal{W}_{\boldsymbol{s}},\ \gcd(\boldsymbol{x}-\boldsymbol{s}) = 1, \qquad \forall \boldsymbol{s} \in S \\
&\iff \sigma(\boldsymbol{x}) \in \mathcal{W} \backslash \bigcup_{\boldsymbol{s}\in S} \mathcal{W}_{\boldsymbol{s}},\ \pi_{\mathfrak{p}}(\boldsymbol{x}) \notin \pi_{\mathfrak{p}}(S), \qquad \forall \mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K} \\
&\iff \boldsymbol{x} \in \Lambda(\mathcal{W}_2, \mathcal{L}_2).
\end{aligned}$$

Since $\mathcal{W}_{\boldsymbol{s}} \subset \mathcal{W}$ and S is finite, by inclusion-exclusion

$$\begin{aligned}
\theta_H(\mathcal{W}_2) &= \mathrm{vol}\left(\mathcal{W} \backslash \left(\bigcup_{\boldsymbol{s}\in S} \mathcal{W}_{\boldsymbol{s}}\right)\right) \cdot \prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right) \\
&= \left(\mathrm{vol}(\mathcal{W}) - \sum_{\varnothing \neq I \subseteq S} (-1)^{\#(I)+1} \mathrm{vol}\left(\cap_{\boldsymbol{s}\in I} \mathcal{W}_{\boldsymbol{s}}\right)\right) \cdot \prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right) \\
&= \left(\mathrm{vol}(\mathcal{W}) - \sum_{\varnothing \neq I \subseteq S} (-1)^{\#(I)+1} \mathrm{vol}\left(\mathcal{W}_I\right)\right) \cdot \prod_{\mathfrak{p}\in\mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right).
\end{aligned}$$

Since $\mathrm{dens}(\mathcal{L}) = \mathrm{dens}(\mathcal{L}_2)$, by Theorem 1.1, we get

$$\theta(\Lambda_{\mathrm{vis},S}) = \theta(\Lambda(\mathcal{W}_2, \mathcal{L}_2)) = \mathrm{dens}(\mathcal{L}_2)\theta_H(\mathcal{W}_2).$$

The assumption $\theta(\Lambda_{\mathrm{vis},S}) > 0$ implies $\theta_H(\mathcal{W}_2) > 0$. Since $\mathcal{W}_2$ is relatively compact and $\theta_H(\mathcal{W}_2) > 0$, the set $\Lambda_{\mathrm{vis},S} = \Lambda(\mathcal{W}_2, \mathcal{L}_2)$ is a weak model set of maximal density.

6. Proof of Theorem 1.3

In this section we will discuss simultaneous visibility in μ-almost every cut-and-project set. At the end we present the proof of Theorem 1.3.

6.1. **Auxiliary results.** Let $n = d + m$, where $d \geqslant 2$ and $m \geqslant 1$. Let $\mathcal{L} \subset \mathbb{R}^n$ be a lattice and let $\mathcal{W}$ be a bounded convex window with nonempty interior. Assume that $\Lambda(\mathcal{W}, \mathcal{L})$ is an irreducible cut-and-project set. Since π_{phys} is injective, for each $\boldsymbol{s} \in S$, we have a unique lift $\tilde{\boldsymbol{s}} \in \mathcal{L}$, such that $\pi_{\mathrm{phys}}(\tilde{\boldsymbol{s}}) = \boldsymbol{s}$. Let

$$\tilde{S} = \{\tilde{\boldsymbol{s}} \in \mathcal{L} \,:\, \pi_{\mathrm{phys}}(\tilde{\boldsymbol{s}}) \in S\}.$$

Define

$$\mathcal{L}_{\mathrm{vis},\tilde{S}} \stackrel{\mathrm{def}}{=} \{\boldsymbol{y} \in \mathcal{L}\backslash\tilde{S} \,:\, \boldsymbol{y} \text{ is visible from every } \tilde{\boldsymbol{s}} \in \tilde{S}\},$$

and

$$\Lambda\Big(\mathcal{W}, \mathcal{L}_{\mathrm{vis},\tilde{S}}\Big) \stackrel{\mathrm{def}}{=} \{\pi_{\mathrm{phys}}(\boldsymbol{y}) \,:\, \boldsymbol{y} \in \mathcal{L}_{\mathrm{vis},\tilde{S}},\ \pi_{\mathrm{int}}(\boldsymbol{y}) \in \mathcal{W}\}.$$

Lemma 6.1. *Let $\mathcal{L}$ be a lattice in $\mathbb{R}^n$, and $\mathcal{W}$ be a bounded convex window with nonempty interior. Then*

$$\Lambda(\mathcal{W}, \mathcal{L})_{\mathrm{vis},S} \subseteq \Lambda\Big(\mathcal{W}, \mathcal{L}_{\mathrm{vis},\tilde{S}}\Big), \tag{29}$$

for every finite set $S \subset \Lambda(\mathcal{W}, \mathcal{L})$.

Proof. Let $\boldsymbol{x} \in \Lambda(\mathcal{W}, \mathcal{L})\backslash S$ be arbitrary. It is enough to show that if $\boldsymbol{x} \notin \Lambda\Big(\mathcal{W}, \mathcal{L}_{\mathrm{vis},\tilde{S}}\Big)$ then $\boldsymbol{x} \notin \Lambda(\mathcal{W}, \mathcal{L})_{\mathrm{vis},S}$. Since $\boldsymbol{x} \in \Lambda(\mathcal{W}, \mathcal{L})\backslash S$, there exists a unique $\boldsymbol{y} \in \mathcal{L}$ and $\boldsymbol{y} \neq \tilde{\boldsymbol{s}}$ for all $\tilde{\boldsymbol{s}} \in \tilde{S}$, such that

$$\boldsymbol{x} = \pi_{\mathrm{phys}}(\boldsymbol{y}) \qquad \text{and} \qquad \pi_{\mathrm{int}}(\boldsymbol{y}) \in \mathcal{W}.$$

The assumption $\boldsymbol{x} \notin \Lambda\Big(\mathcal{W}, \mathcal{L}_{\mathrm{vis},\tilde{S}}\Big)$ implies that $\boldsymbol{y} \notin \mathcal{L}_{\mathrm{vis},\tilde{S}}$. Hence there exists $\tilde{\boldsymbol{s}}_0 \in \tilde{S}$, $\tilde{\boldsymbol{s}}_0 \neq \boldsymbol{y}$ and $t \in (0,1)$ such that

$$\boldsymbol{y}' = t\boldsymbol{y} + (1-t)\tilde{\boldsymbol{s}}_0 \in \mathcal{L}.$$

Since $\pi_{\mathrm{int}}(\boldsymbol{y}), \pi_{\mathrm{int}}(\tilde{\boldsymbol{s}}_0) \in \mathcal{W}$, and the window $\mathcal{W}$ is convex, we have

$$\pi_{\mathrm{int}}(\boldsymbol{y}') = t\pi_{\mathrm{int}}(\boldsymbol{y}) + (1-t)\pi_{\mathrm{int}}(\tilde{\boldsymbol{s}}_0) \in \mathcal{W}.$$

Therefore

$$\pi_{\mathrm{phys}}(\boldsymbol{y}') \in \Lambda(\mathcal{W}, \mathcal{L}),$$

and

$$\pi_{\mathrm{phys}}(\boldsymbol{y}') = t\pi_{\mathrm{phys}}(\boldsymbol{y}) + (1-t)\pi_{\mathrm{phys}}(\tilde{\boldsymbol{s}}_0) = t\boldsymbol{x} + (1-t)\boldsymbol{s}_0.$$

Since $0 < t < 1$, the point $\pi_{\mathrm{phys}}(\boldsymbol{y}')$ lies in the open line segment $(\boldsymbol{s}_0, \boldsymbol{x})$. Hence

$$\boldsymbol{x} \notin \Lambda(\mathcal{W}, \mathcal{L})_{\mathrm{vis},S}.$$

□

Lemma 6.2. *Let $\mathcal{L} \subset \mathbb{R}^{d+m}$ be a lattice, where $d \geqslant 2$, $m \geqslant 1$, and let $\mathcal{W} \subset \mathbb{R}^m$ be a bounded convex window with nonempty interior. If (29) is a strict inclusion,*

$$\Lambda(\mathcal{W}, \mathcal{L})_{\mathrm{vis},S} \subsetneq \Lambda\Big(\mathcal{W}, \mathcal{L}_{\mathrm{vis},\tilde{S}}\Big),$$

then there exist linearly independent vectors $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathcal{L}$ such that

$$\dim\left(\mathrm{span}_{\mathbb{R}}(\boldsymbol{v}_1, \boldsymbol{v}_2) \cap (\{\boldsymbol{0}\} \times \mathbb{R}^m)\right) = 1.$$

Lemma 6.3. *Let $n = d + m$ for positive integers $d \geqslant 2$ and $m \geqslant 1$. Suppose the window $\mathcal{W} \subset \mathbb{R}^m$ is a bounded convex window with nonempty interior. Let* m *denote the Haar measure on* $\mathrm{GL}_n(\mathbb{R})$. *Then for* m*-a.e. $g \in \mathrm{GL}_n(\mathbb{R})$, we have*

$$\Lambda(\mathcal{W}, (g\mathbb{Z}^n)_{\mathrm{vis},\tilde{S}}) = \Lambda(\mathcal{W}, g\mathbb{Z}^n)_{\mathrm{vis},S}.$$

For every finite set $S \subset \Lambda(\mathcal{W}, g\mathbb{Z}^n)$ and $\tilde{S} \overset{\mathrm{def}}{=} (\pi_{\mathrm{phys}}|_{g\mathbb{Z}^n})^{-1}(S)$. In particular, for μ-a.e. cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$, we have

$$\Lambda(\mathcal{W}, \mathcal{L}_{\mathrm{vis},\tilde{S}}) = \Lambda(\mathcal{W}, \mathcal{L})_{\mathrm{vis},S}.$$

Proof. The proof is analogous to [GKW25, Proposition 3.3] in combination with Lemma 6.2. □

Proof of Lemma 6.2. Assume that the inclusion in (29) is strict. Thus, there exists a point

$$\boldsymbol{x}_1 \in \Lambda\Big(\mathcal{W}, \mathcal{L}_{\mathrm{vis},\tilde{S}}\Big) \backslash \Lambda(\mathcal{W}, \mathcal{L})_{\mathrm{vis},S},$$

such that $\boldsymbol{x}_1 \notin S$. By definition of $\Lambda\Big(\mathcal{W}, \mathcal{L}_{\mathrm{vis},\tilde{S}}\Big)$, there exists $\boldsymbol{y}_1 \in \mathcal{L}_{\mathrm{vis},\tilde{S}}$ such that

$$\pi_{\mathrm{phys}}(\boldsymbol{y}_1) = \boldsymbol{x}_1, \qquad \pi_{\mathrm{int}}(\boldsymbol{y}_1) \in \mathcal{W}.$$

Since $\boldsymbol{y}_1 \in \mathcal{L}_{\mathrm{vis},\tilde{S}}$, the point $\boldsymbol{y}_1$ is visible from every point $\tilde{\boldsymbol{s}} \in \tilde{S}$. On the other hand, $\boldsymbol{x}_1 \notin \Lambda(\mathcal{W}, \mathcal{L})_{\mathrm{vis},S}$, so there exist $\boldsymbol{s}_0 \in S$ and $t \in (0,1)$ such that

$$\boldsymbol{x}_2 \overset{\mathrm{def}}{=} t\boldsymbol{x}_1 + (1-t)\boldsymbol{s}_0 \in \Lambda(\mathcal{W}, \mathcal{L}). \tag{30}$$

Since $\boldsymbol{x}_2 \in \Lambda(\mathcal{W}, \mathcal{L})$, there exists $\boldsymbol{y}_2 \in \mathcal{L}$, such that $\pi_{\mathrm{phys}}(\boldsymbol{y}_2) = \boldsymbol{x}_2$. By (30), we have

$$\pi_{\mathrm{phys}}(\boldsymbol{y}_2 - \tilde{\boldsymbol{s}}_0) = t\pi_{\mathrm{phys}}(\boldsymbol{y}_1 - \tilde{\boldsymbol{s}}_0).$$

Define

$$\boldsymbol{v}_1 \overset{\mathrm{def}}{=} \boldsymbol{y}_1 - \tilde{\boldsymbol{s}}_0, \qquad \boldsymbol{v}_2 \overset{\mathrm{def}}{=} \boldsymbol{y}_2 - \tilde{\boldsymbol{s}}_0.$$

Thus, we have $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathcal{L}$ and $\pi_{\mathrm{phys}}(\boldsymbol{v}_2) = t\pi_{\mathrm{phys}}(\boldsymbol{v}_1)$. Since $\boldsymbol{x}_1 \neq \boldsymbol{s}_0$, we have

$$\pi_{\mathrm{phys}}(\boldsymbol{v}_1) = \boldsymbol{x}_1 - \boldsymbol{s}_0 \neq \boldsymbol{0}. \tag{31}$$

Hence $\boldsymbol{v}_1 \neq \mathbf{0}$ and $\boldsymbol{v}_2 \neq \mathbf{0}$. The vectors $\boldsymbol{v}_1, \boldsymbol{v}_2$ are linearly independent. In fact, if $\boldsymbol{v}_2 = c\boldsymbol{v}_1$ for some $c \in \mathbb{R}$, applying π_{phys} gives $c = t$ and $\boldsymbol{v}_2 = t\boldsymbol{v}_1$, implying

$$\begin{aligned} \boldsymbol{y}_2 - \tilde{\boldsymbol{s}}_0 &= t(\boldsymbol{y}_1 - \tilde{\boldsymbol{s}}_0) \\ \boldsymbol{y}_2 &= t\boldsymbol{y}_1 + (1-t)\tilde{\boldsymbol{s}}_0, \end{aligned}$$

which contradicts that $\boldsymbol{y}_1 \in \mathcal{L}_{\text{vis},\tilde{S}}$.

Therefore, letting

$$U \stackrel{\text{def}}{=} \operatorname{span}_{\mathbb{R}}(\boldsymbol{v}_1, \boldsymbol{v}_2) \subset \mathbb{R}^{d+m},$$

we have that

$$\dim U = 2.$$

By (31) we obtain that $\dim(\pi_{\text{phys}}(U)) = 1$, and since the kernel of π_{phys} is the space $\{\mathbf{0}\} \times \mathbb{R}^m$ then,

$$\dim(\operatorname{span}_{\mathbb{R}}(\boldsymbol{v}_1, \boldsymbol{v}_2) \cap (\{\mathbf{0}\} \times \mathbb{R}^m)) = \dim \ker(\pi_{\text{phys}}|_U) = 1.$$

□

Lemma 6.4. *Let $n = d + m$ for positive integers $d \geqslant 2$, $m \geqslant 1$, and $\mathcal{L} \subset \mathbb{R}^n$ be a lattice. Let $\tilde{S} \subset \mathcal{L}$ be a finite set and let $\mathcal{W} \subset \mathbb{R}^m$ be a bounded convex window of positive Lebesgue measure. Suppose that $\Lambda(\mathcal{W}, \mathcal{L})$ is an irreducible cut-and-project set. Then the density of $\Lambda\Big(\mathcal{W}, \mathcal{L}_{\text{vis},\tilde{S}}\Big)$ along $\{TD\}_{T>0}$ exists and it is equal to*

$$\theta\Big(\Lambda\Big(\mathcal{W}, \mathcal{L}_{\text{vis},\tilde{S}}\Big)\Big) = \frac{\operatorname{vol}(\mathcal{W})}{\operatorname{covol}(\mathcal{L})} \prod_{p \in \mathcal{P}} \left(1 - \frac{s_{\mathcal{L}}(p)}{p^n}\right),$$

where D is a bounded convex set with positive measure.

Proof. Consider the groups $G = \mathbb{R}^d$, and $H = \mathbb{R}^m \times \prod_{p \in \mathcal{P}} H_p$, where

$$H_p = \mathcal{L}/p\mathcal{L}\,, \qquad p \in \mathcal{P}.$$

In the group H, each finite factor is equipped with the normalized counting measure. Both G and H are σ-compact and LCAGs, and the group H is compactly generated. Let $\mathcal{L}'$ be the image of the diagonal embedding

$$\mathcal{L} \hookrightarrow G \times H, \qquad \boldsymbol{y} \in \mathcal{L} \longmapsto (\boldsymbol{y}, (\pi_{\mathcal{L},p}(\boldsymbol{y}))_{p \in \mathcal{P}}).$$

The restriction $\pi'_{\text{phys}}|_{\mathcal{L}'} : \mathcal{L}' \to G$ is injective. Using the fact that $\pi_{\text{int}}(\mathcal{L})$ is dense in $\mathbb{R}^m$ and $H_p \cong (\mathbb{Z}\backslash p\mathbb{Z})^n$, it can be shown that $\pi'_{\text{int}}(\mathcal{L}')$ is dense in H. Furthermore, $\mathcal{L}'$ is a cocompact lattice. Define

$$\mathcal{W}_p \stackrel{\text{def}}{=} (\mathcal{L}/p\mathcal{L})\backslash\pi_{\mathcal{L},p}(\tilde{S}).$$

and

$$\mathcal{W}' \stackrel{\text{def}}{=} \mathcal{W} \times \prod_{p \in \mathcal{P}} \mathcal{W}_p.$$

By construction, the associated cut-and-project set $\Lambda(\mathcal{W}', \mathcal{L}')$ consists exactly of those $\boldsymbol{x} \in \mathbb{R}^d$ such that there exists unique $\boldsymbol{y} \in \mathcal{L}$

$$\pi_{\text{phys}}(\boldsymbol{y}) = \boldsymbol{x}, \qquad \pi_{\text{int}}(\boldsymbol{y}) \in \mathcal{W}, \qquad \pi_{\mathcal{L},p}(\boldsymbol{y}) \notin \pi_{\mathcal{L},p}(\tilde{S}), \qquad p \in \mathcal{P}.$$

Equivalently,

$$\Lambda(\mathcal{W}, \mathcal{L}_{\text{vis},\tilde{S}}) = \Lambda(\mathcal{W}', \mathcal{L}').$$

If $s_{\mathcal{L}}(p) = p^n$ for some prime p, then $\mathcal{W}_p = \varnothing$, consequently $\mathcal{W}' = \varnothing$. Therefore, the lemma is immediate. In the rest of the proof, we assume that

$$s_{\mathcal{L}}(p) \neq p^n, \qquad p \in \mathcal{P}.$$

Let p_i be the i-th rational prime. Define

$$\mathcal{W}'_t = \mathcal{W} \times \prod_{i=1}^{t} \mathcal{W}_{p_i} \times \prod_{i>t} H_{p_i}, \qquad t \in \mathbb{N}.$$

For fixed $t \in \mathbb{N}$, the cut-and-project set

$$\Lambda(\mathcal{W}'_t, \mathcal{L}'),$$

is an irreducible cut-and-project set. The density is given by

$$\theta(\Lambda(\mathcal{W}'_t, \mathcal{L}')) = \frac{\mathrm{vol}(\mathcal{W}'_t)}{\mathrm{covol}(\mathcal{L}')} = \frac{\mathrm{vol}(\mathcal{W})}{\mathrm{covol}(\mathcal{L})} \cdot \prod_{i=1}^{t} \left(1 - \frac{s_{\mathcal{L}}(p_i)}{p_i^n}\right). \tag{32}$$

Since

$$\Lambda(\mathcal{W}', \mathcal{L}') \subseteq \Lambda(\mathcal{W}'_t, \mathcal{L}') \tag{33}$$

for every $t \in \mathbb{N}$. By (32), we get

$$\overline{\theta}(\Lambda(\mathcal{W}', \mathcal{L}')) \leqslant \frac{\mathrm{vol}(\mathcal{W})}{\mathrm{covol}(\mathcal{L})} \cdot \prod_{i=1}^{t} \left(1 - \frac{s_{\mathcal{L}}(p_i)}{p_i^n}\right).$$

Letting $t \to \infty$, we obtain

$$\overline{\theta}(\Lambda(\mathcal{W}', \mathcal{L}')) \leqslant \frac{\mathrm{vol}(\mathcal{W})}{\mathrm{covol}(\mathcal{L})} \cdot \prod_{i=1}^{\infty} \left(1 - \frac{s_{\mathcal{L}}(p_i)}{p_i^n}\right).$$

Define

$$A_t \stackrel{\mathrm{def}}{=} \Lambda(\mathcal{W}'_t, \mathcal{L}') \backslash \Lambda(\mathcal{W}', \mathcal{L}'). \tag{34}$$

By eqs. (32) and (33), the sets $B = \Lambda(\mathcal{W}', \mathcal{L}')$ and $A = \Lambda(\mathcal{W}'_t, \mathcal{L}')$ satisfy the hypothesis of Lemma 4.6. Therefore

$$\underline{\theta}(\Lambda(\mathcal{W}', \mathcal{L}')) = \theta(\Lambda(\mathcal{W}'_t, \mathcal{L}')) - \overline{\theta}(A_t). \tag{35}$$

We state the following, postpone its proof until the end, and finish the proof of Lemma 6.4.

Lemma 6.5. *Let A_t be as in (34). Then $\lim_{t\to\infty} \overline{\theta}(A_t) = 0$.*

Combining the previous lemma with (35) and taking the limit as $t \to \infty$,

$$\overline{\theta}(\Lambda(\mathcal{W}', \mathcal{L}')) \leqslant \frac{\mathrm{vol}(\mathcal{W})}{\mathrm{covol}(\mathcal{L})} \cdot \prod_{i=1}^{\infty} \left(1 - \frac{s_{\mathcal{L}}(p_i)}{p_i^n}\right) = \underline{\theta}(\Lambda(\mathcal{W}', \mathcal{L}')).$$

This proves that the density $\theta(\Lambda(\mathcal{W}, \mathcal{L}_{\mathrm{vis},\tilde{S}}))$ along $\{TD\}_{T>0}$ exists and is equal

$$\frac{\mathrm{vol}(\mathcal{W})}{\mathrm{covol}(\mathcal{L})} \cdot \prod_{p\in\mathcal{P}} \left(1 - \frac{s_{\mathcal{L}}(p)}{p^n}\right)$$

establishing Lemma 6.4. □

Proof of Lemma 6.5. We claim that

$$A_t \subseteq \bigcup_{\boldsymbol{s}\in S} \bigcup_{\substack{p\in\mathcal{P} \\ p>p_t}} (\boldsymbol{s} + \Lambda(\mathcal{W} - \pi_{\mathrm{int}}(\tilde{\boldsymbol{s}}), p\mathcal{L}) \backslash \{\boldsymbol{s}\}), \tag{36}$$

where $\tilde{\boldsymbol{s}} = (\pi_{\mathrm{phys}}|_{\mathcal{L}})^{-1}(\boldsymbol{s})$ is the unique lift of $\boldsymbol{s} \in \Lambda$ to $\mathcal{L}$. Indeed, let $\boldsymbol{x} \in A_t$ be an arbitrary element. Since $\boldsymbol{x} \in \Lambda(\mathcal{W}'_t, \mathcal{L}')$, there is unique $\boldsymbol{y} \in \mathcal{L}$, such that $\boldsymbol{y} \neq \tilde{\boldsymbol{s}}$ for all $\tilde{\boldsymbol{s}} \in \tilde{S}$ and

$$\pi_{\mathrm{phys}}(\boldsymbol{y}) = \boldsymbol{x}, \quad \pi_{\mathrm{int}}(\boldsymbol{y}) \in \mathcal{W},$$

and

$$\pi_{\mathcal{L},p}(\boldsymbol{y}) \notin \pi_{\mathcal{L},p}(\tilde{S}), \qquad p \in \mathcal{P},\ p \leqslant p_t.$$

On the other hand, $\boldsymbol{x} \notin \Lambda(\mathcal{W}', \mathcal{L}')$ implies that there is a prime $p > p_t$ and $\tilde{\boldsymbol{s}}_0 \in \tilde{S}$, such that

$$\boldsymbol{y} \equiv \tilde{\boldsymbol{s}}_0 \mod p\mathcal{L}.$$

Hence

$$\boldsymbol{x} \in \pi_{\text{phys}}(\tilde{\boldsymbol{s}}_0) + \Lambda(\mathcal{W} - \pi_{\text{int}}(\tilde{\boldsymbol{s}}_0), p\mathcal{L}), \qquad \text{and} \qquad \boldsymbol{x} \neq \boldsymbol{s}_0.$$

This completes the proof of the claim.

For $\boldsymbol{s} \in S$ and $p \in \mathcal{P}$, write

$$N_{\boldsymbol{s},p,T} \stackrel{\text{def}}{=} \#(\boldsymbol{s} + \Lambda(\mathcal{W} - \pi_{\text{int}}(\tilde{\boldsymbol{s}}), p\mathcal{L}) \backslash \{\boldsymbol{s}\} \cap TD)\,.$$

Let $\lambda_1(\mathcal{L}) = \min\{|\boldsymbol{v}| : \boldsymbol{v} \in \mathcal{L}\backslash\{\boldsymbol{0}\}\} > 0$ be the first successive minima. Choose constants $R_D, R_S, R_{\mathcal{W}} > 0$ such that

$$TD \subseteq B^d_{R_D T}(\boldsymbol{0}) \quad \text{for all } T > 0, \qquad |\boldsymbol{s}| \leqslant R_S \quad \text{for all } \boldsymbol{s} \in S,$$

and

$$|\boldsymbol{u} - \pi_{\text{int}}(\tilde{\boldsymbol{s}})| \leqslant R_{\mathcal{W}} \quad \text{for all } \boldsymbol{s} \in S \text{ and all } \boldsymbol{u} \in \mathcal{W}.$$

Suppose that $N_{\boldsymbol{s},p,T} > 0$. Then there exist $\boldsymbol{s} \neq \boldsymbol{x} \in TD$, $\boldsymbol{u} \in \mathcal{W}$ and $\boldsymbol{v} \in \mathcal{L}\backslash\{\boldsymbol{0}\}$ such that

$$\boldsymbol{x} - \boldsymbol{s} = p\pi_{\text{phys}}(\boldsymbol{v}), \qquad \boldsymbol{u} - \pi_{\text{int}}(\tilde{\boldsymbol{s}}) = p\pi_{\text{int}}(\boldsymbol{v}).$$

Thus,

$$p\boldsymbol{v} = (\boldsymbol{x} - \boldsymbol{s}, \boldsymbol{u} - \pi_{\text{int}}(\tilde{\boldsymbol{s}})).$$

It follows that

$$\begin{aligned} p\lambda_1(\mathcal{L}) \leqslant |p\boldsymbol{v}| = |(\boldsymbol{x} - \boldsymbol{s}, \boldsymbol{u} - \pi_{\text{int}}(\tilde{\boldsymbol{s}}))| &\leqslant |\boldsymbol{x} - \boldsymbol{s}| + |\boldsymbol{u} - \pi_{\text{int}}(\tilde{\boldsymbol{s}})| \\ &\leqslant |\boldsymbol{x}| + |\boldsymbol{s}| + R_{\mathcal{W}} \leqslant R_D T + R_S + R_{\mathcal{W}}. \end{aligned}$$

For $T > 1$, we have $R_D T + R_S + R_{\mathcal{W}} \leqslant T(R_D + R_S + R_{\mathcal{W}})$. Hence

$$p \leqslant aT, \qquad a = \frac{R_D + R_S + R_{\mathcal{W}}}{\lambda_1(\mathcal{L})}.$$

Since the constant a is independent of $\boldsymbol{s}, p, t, T$, there are only finitely many primes p in (36) that contribute after intersecting with TD.

Every point counted by $N_{\boldsymbol{s},p,T}$ corresponds to a point $\boldsymbol{v} \in \mathcal{L}\backslash\{\boldsymbol{0}\}$ satisfying

$$\pi_{\text{phys}}(\boldsymbol{v}) \in \frac{TD - \boldsymbol{s}}{p}, \qquad \pi_{\text{int}}(\boldsymbol{v}) \in \frac{\mathcal{W} - \pi_{\text{int}}(\tilde{\boldsymbol{s}})}{p}.$$

Note that

$$\frac{\mathcal{W} - \pi_{\text{int}}(\tilde{\boldsymbol{s}})}{p} \subseteq B_{R_{\mathcal{W}}}(\boldsymbol{0}), \qquad \boldsymbol{s} \in S,\, p \geqslant 2,$$

and

$$\frac{TD - \boldsymbol{s}}{p} \subseteq B_{\frac{(R_D + R_S)T}{p}}(\boldsymbol{0}), \qquad \boldsymbol{s} \in S,\, p \geqslant 2,\, T > 1.$$

To simplify the notation, we denote

$$B^{T,p} = B_{\frac{(R_D + R_S)T}{p}}(\boldsymbol{0}), \qquad T > 1,$$

and

$$B_{\mathcal{W}}(\boldsymbol{0}) = B_{\mathcal{W}}.$$

We have

$$N_{\boldsymbol{s},p,T} \leqslant \#\left(\Lambda(B_{\mathcal{W}}, \mathcal{L}) \cap B^{T,p}\right), \qquad \boldsymbol{s} \in S,\, T > 1,\, p \in \mathcal{P}$$

Therefore, for sufficiently large T, by (36), we have

$$\begin{aligned} \#(A_t \cap TD) &\leqslant \sum_{\boldsymbol{s}\in S} \sum_{p_t<p\leqslant aT} N_{\boldsymbol{s},p,T} \\ &\leqslant \sum_{\boldsymbol{s}\in S} \sum_{p_t<p\leqslant aT} \#\big(\Lambda(B_{\mathcal{W}},\mathcal{L}) \cap B^{T,p}\big) \end{aligned} \tag{37}$$

Since $\Lambda(B_{\mathcal{W}},\mathcal{L})$ is uniformly discrete, there exists a $\delta > 0$ such that

$$|\boldsymbol{x}-\boldsymbol{y}| > \delta, \qquad \boldsymbol{x} \neq \boldsymbol{y} \in \Lambda(B_{\mathcal{W}},\mathcal{L}).$$

Consequently, there exists a constant $C_{\Lambda(B_{\mathcal{W}},\mathcal{L})}$ independent of $\boldsymbol{s}, p, t, T$ such that

$$\begin{aligned} \#\big(\Lambda(B_{\mathcal{W}},\mathcal{L}) \cap B^{T,p}\big) &\leqslant C_{\Lambda(B_{\mathcal{W}},\mathcal{L})} \left(1 + \frac{2(R_D+R_S)T}{p}\right)^d \\ &\leqslant C'_{\Lambda(B_{\mathcal{W}},\mathcal{L})} \left(1+\frac{T}{p}\right)^d, \end{aligned} \qquad T>1,\ p\in\mathcal{P} \tag{38}$$

Using the fact that $p \leqslant aT$, we get

$$\#\big(\Lambda(B_{\mathcal{W}},\mathcal{L}) \cap B^{T,p}\big) \leqslant C''_{\Lambda(B_{\mathcal{W}},\mathcal{L})} \left(\frac{T}{p}\right)^d, \qquad T>1,\ p\in\mathcal{P}, p_t<p\leqslant aT, \tag{39}$$

where the constant $C''_{\Lambda(B_{\mathcal{W}},\mathcal{L})}$ is independent of $\boldsymbol{s}, p, t, T$. Since S is finite, we get

$$\begin{aligned} \#(A_t \cap TD) &\leqslant \#(S)\, C''_{\Lambda(B_{\mathcal{W}},\mathcal{L})} \sum_{p_t<p\leqslant aT} \left(\frac{T}{p}\right)^d \\ \frac{\#(A_t \cap TD)}{\mathrm{vol}(TD)} &\leqslant \#(S)\, C'''_{\Lambda(B_{\mathcal{W}},\mathcal{L})} \sum_{p_t<p\leqslant aT} \frac{1}{p^d} \\ \frac{\#(A_t \cap TD)}{\mathrm{vol}(TD)} &\leqslant \#(S)\, C'''_{\Lambda(B_{\mathcal{W}},\mathcal{L})} \sum_{p_t<p} \frac{1}{p^d}, \end{aligned} \tag{40}$$

where $C'''_{\Lambda(B_{\mathcal{W}},\mathcal{L})}$ is a constant independent of t, T. Taking $\limsup_{T\to\infty}$ gives

$$\overline{\theta}(A_t) \leqslant \#(S)\, C'''_{\Lambda(B_{\mathcal{W}},\mathcal{L})} \sum_{p_t<p} \frac{1}{p^d}. \tag{41}$$

Since $d \geqslant 2$,

$$\sum_{p\in\mathcal{P}} \frac{1}{p^2} < \infty$$

taking $t \to \infty$ in (41), we get

$$\lim_{t\to\infty} \overline{\theta}(A_t) = 0.$$

□

7. Proof of Proposition 1.5

Recall that $\overline{\mu}$ is invariant under translations by vectors in $\mathbb{R}^d$ and assigns full measure to the collection of cut-and-project sets $\Lambda(\mathcal{W},\mathcal{L})$ for $\mathcal{L} \in \mathscr{A}_n$. The proof follows from the same ideas as the proof of [GKW25, Proposition 3.1]. For $\overline{\mu}$-a.e. cut-and-project set $\Lambda(\mathcal{W},\mathcal{L})$ and $\boldsymbol{s} \in S$, we have

$$\Lambda(\mathcal{W},\mathcal{L})_{\mathrm{vis},\{\boldsymbol{s}\}} = \Lambda(\mathcal{W},\mathcal{L})\backslash\{\boldsymbol{s}\}.$$

By the definition of visibility from the set S, we have

$$\Lambda(\mathcal{W},\mathcal{L})_{\mathrm{vis},S} = \cap_{\boldsymbol{s}\in S}\Lambda(\mathcal{W},\mathcal{L})_{\mathrm{vis},\{\boldsymbol{s}\}}.$$

Since S is countable, for $\overline{\mu}$-a.e. cut-and-project set $\Lambda(\mathcal{W},\mathcal{L})$

$$\Lambda(\mathcal{W},\mathcal{L})_{\mathrm{vis},S} = \cap_{\boldsymbol{s}\in S}\Lambda(\mathcal{W},\mathcal{L})_{\mathrm{vis},\{\boldsymbol{s}\}} = \cap_{\boldsymbol{s}\in S}\Lambda(\mathcal{W},\mathcal{L})\backslash\{\boldsymbol{s}\} = \Lambda(\mathcal{W},\mathcal{L})\backslash S.$$

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